模拟电子技术基础

Analog Device

5 Frequency response of amplifier circuits 放大电路的频率响应

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5.0 Fundamentals of complex numbers

A, *B*, and *C* are complex numbers复数: $A = \frac{B}{C}$

The amplitude of *A*: $|A| = \frac{|B|}{|C|}$

A can also be expressed as: $A = |A|e^{j\varphi}$

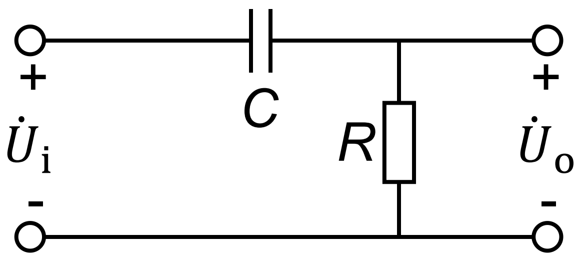
If *A*=*a*+j*b* (*a* is real number 实数 and j*b* 虚数 is imaginary number), *A* can also be expressed as:

Amplitude: $|A| = \sqrt{a^2 + b^2}$

Phase: $\varphi = \arctan(\frac{b}{a})$

5.1 Fundamental of frequency response

High frequency pass circuit 高通电路



$$\dot{U}_\mathrm{o} = \frac{R}{R + \frac{1}{j\omega C}} \dot{U}_\mathrm{i}$$

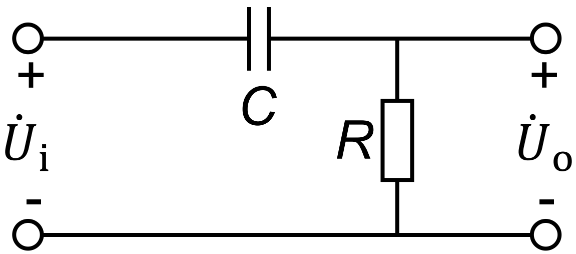
$$\dot{A}_\mathrm{u} = \frac{\dot{U}_\mathrm{o}}{\dot{U}_\mathrm{i}} = \frac{1}{1 + \frac{1}{j\omega RC}} = \frac{1}{1 + \frac{\omega_\mathrm{L}}{j\omega}} \quad , \omega_\mathrm{L} = \frac{\mathbf{1}}{RC}$$

$$\dot{A}_\mathrm{u} = |\dot{A}_\mathrm{u}| e^{j\varphi}$$

$$\dot{A}_\mathrm{u} = \frac{\frac{f^2}{{f_\mathrm{L}}^2}}{1 + \frac{f^2}{{f_\mathrm{L}}^2}} + j\frac{\frac{f}{f_\mathrm{L}}}{1 + \frac{f^2}{{f_\mathrm{L}}^2}} \quad , f = \frac{\omega}{2\pi}, \; f_\mathrm{L} = \frac{\omega_\mathrm{L}}{2\pi}$$

$$\begin{cases} |\dot{A}_\mathrm{u}| = \dfrac{\frac{f}{f_\mathrm{L}}}{\sqrt{1 + \frac{f^2}{{f_\mathrm{L}}^2}}} \\ \varphi = 90^\mathrm{o} - \arctan\dfrac{f}{f_\mathrm{L}} \end{cases}$$

High frequency pass circuit 高通电路



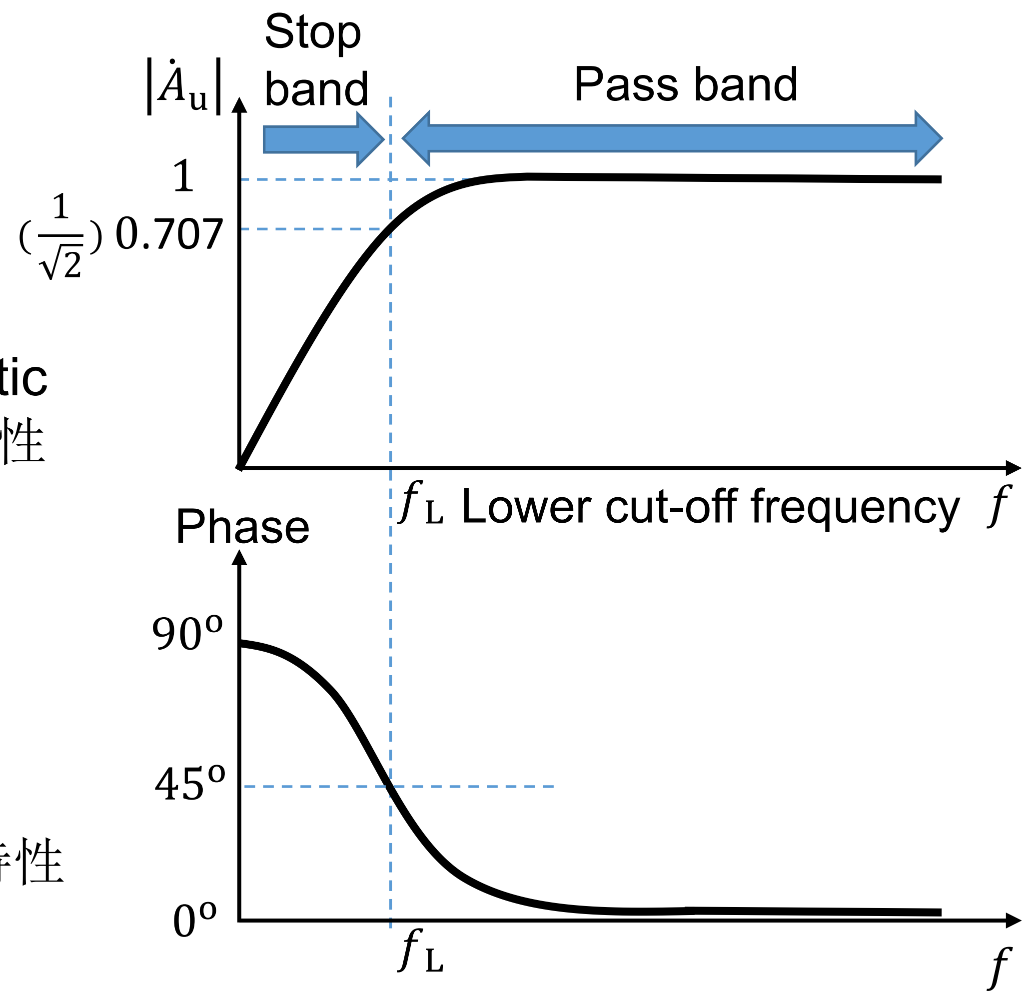
$$\dot{A}_{\mathrm{u}} = \left|\dot{A}_{\mathrm{u}}\right| e^{j\varphi}$$

Amplitude frequency characteristic 幅频特性

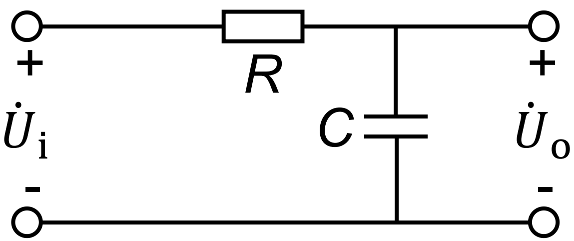
$$\left|\dot{A}_{\mathrm{u}}\right| = \frac{\dfrac{f}{f_{\mathrm{L}}}}{\sqrt{1+\dfrac{f^2}{{f_{\mathrm{L}}}^2}}}$$

Phase frequency characteristic 相频特性

$$\varphi = 90^{\mathrm{o}} - \arctan\frac{f}{f_{\mathrm{L}}}$$



Low frequency pass circuit 低通电路



$$\dot{U}_{\mathrm{o}} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \dot{U}_{\mathrm{i}}$$

$$\dot{A}_{\mathrm{u}} = \left|\dot{A}_{\mathrm{u}}\right| e^{i\varphi}$$

Amplitude frequency characteristic

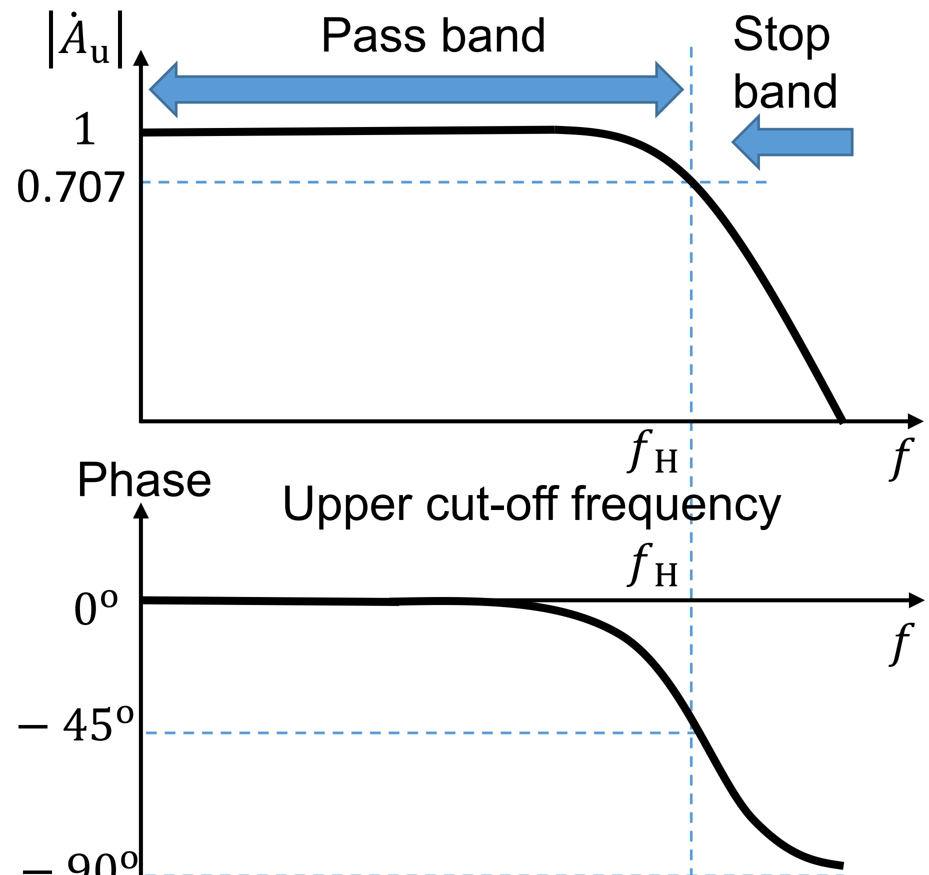
幅频特性

$$\left|\dot{A}_{\mathrm{u}}\right| = \frac{1}{\sqrt{1 + \frac{f^2}{{f_{\mathrm{H}}}^2}}}, f_{\mathrm{H}} = \frac{1}{2\pi RC}$$

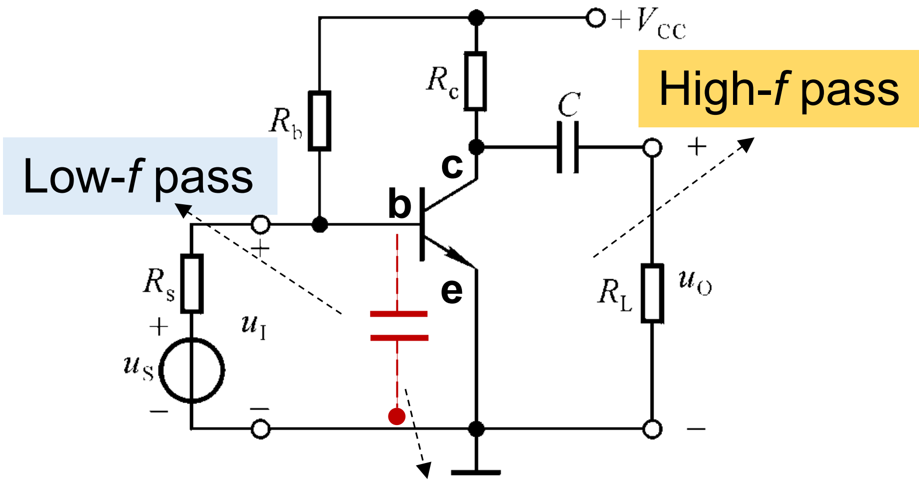
Phase frequency characteristic

相频特性

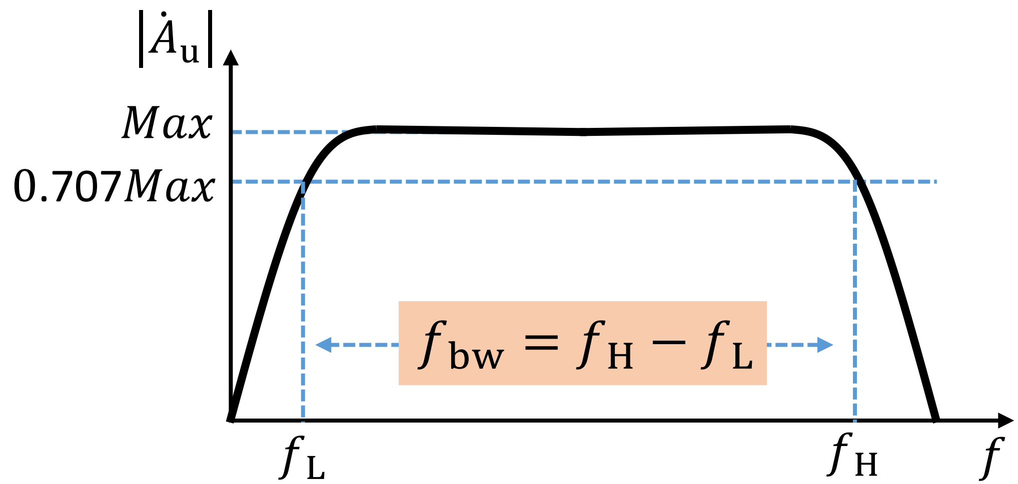
$$\varphi = -\arctan\frac{f}{f_{\mathrm{H}}}$$



Bandwidth in amplifier circuit



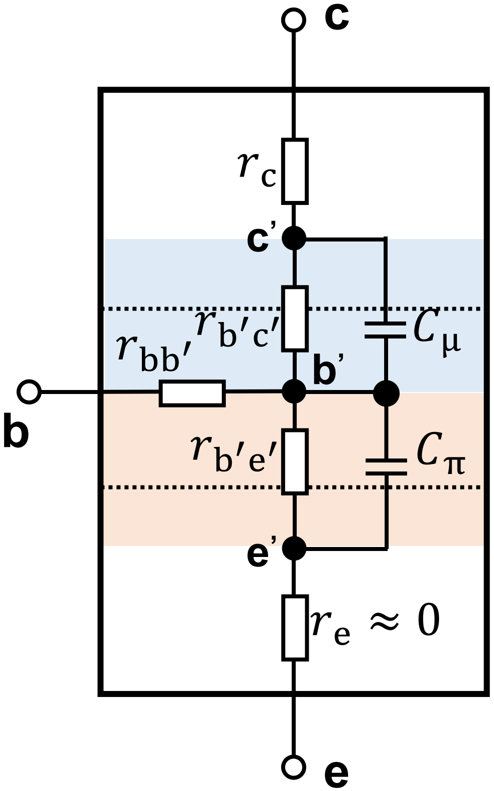
BE junction capacitor 结电容

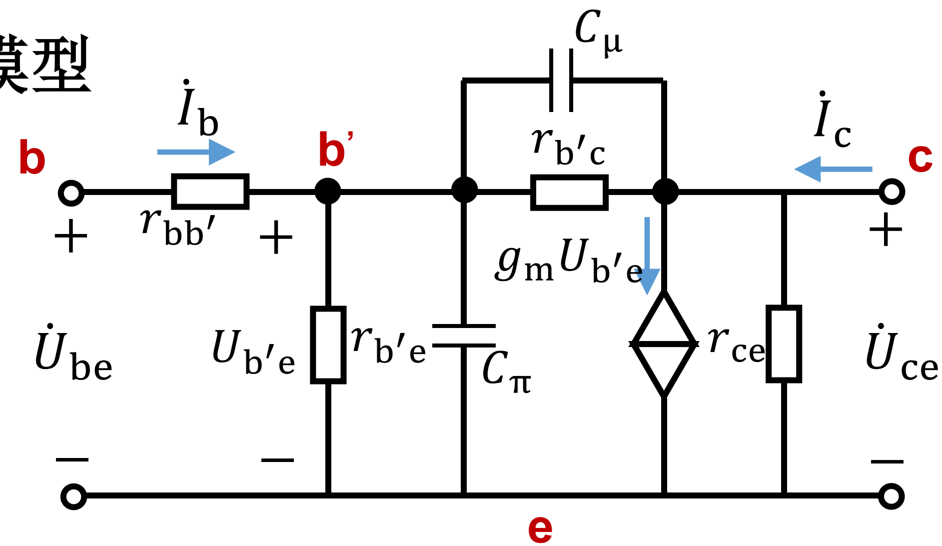


f_{bw}：bandwidth 通频带、带宽

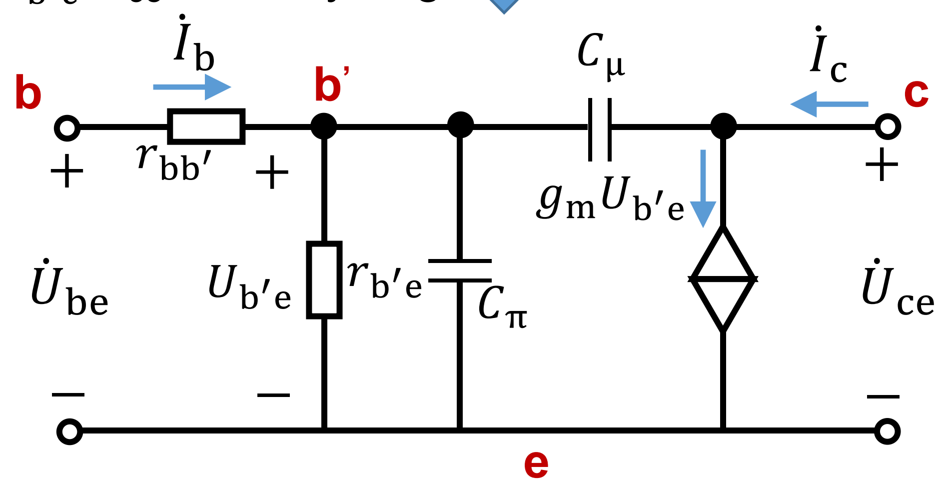
5.2 The high frequency model of transistor

Hybrid - π model 混合π模型

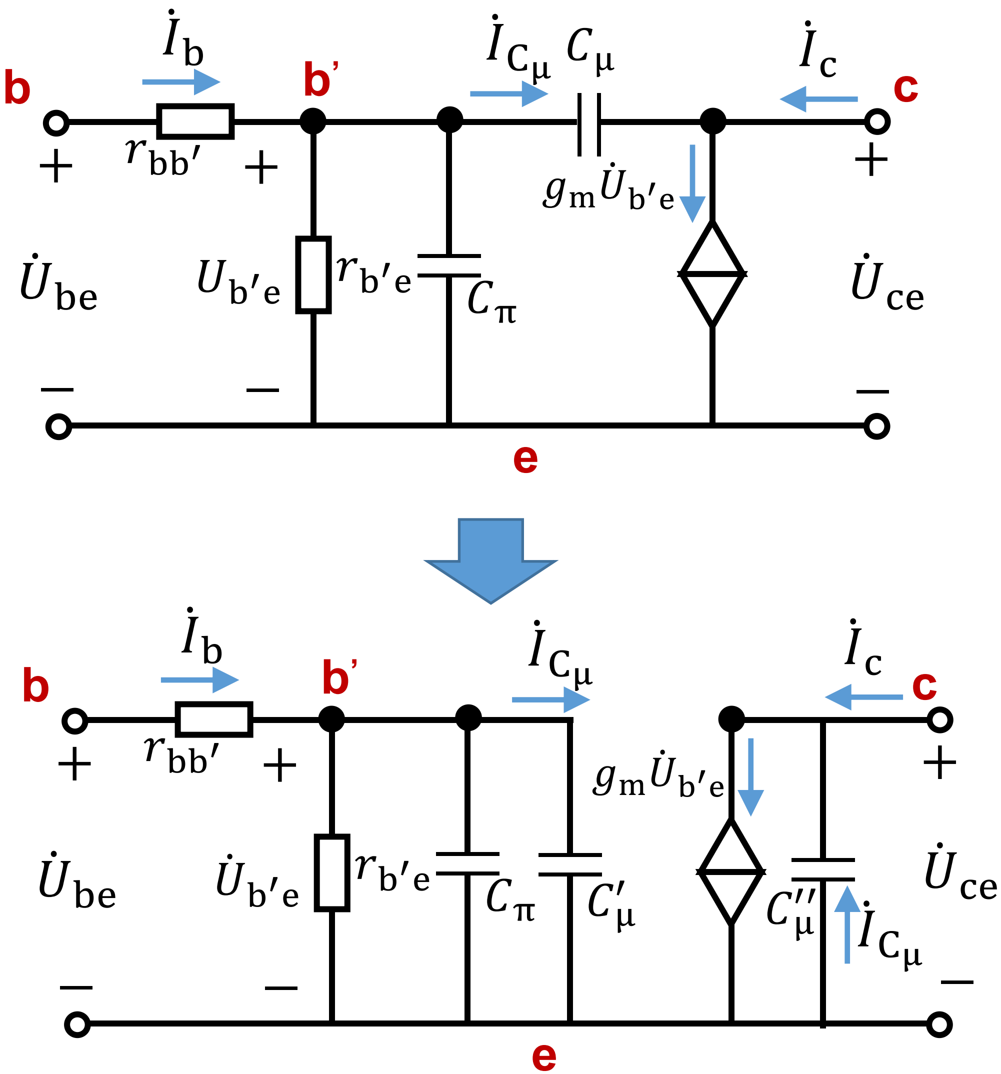




$r_{b'c}$, r_{ce} are very large



Unilateralization of hybrid - π model 混合π模型单向化

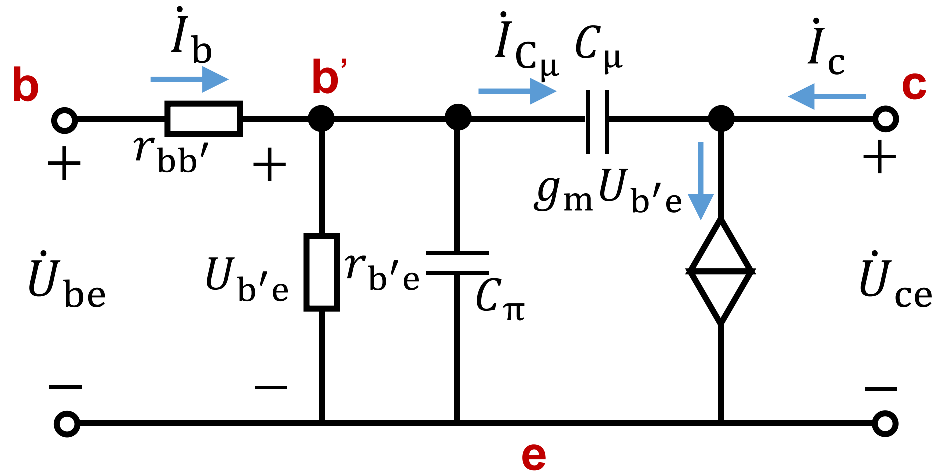


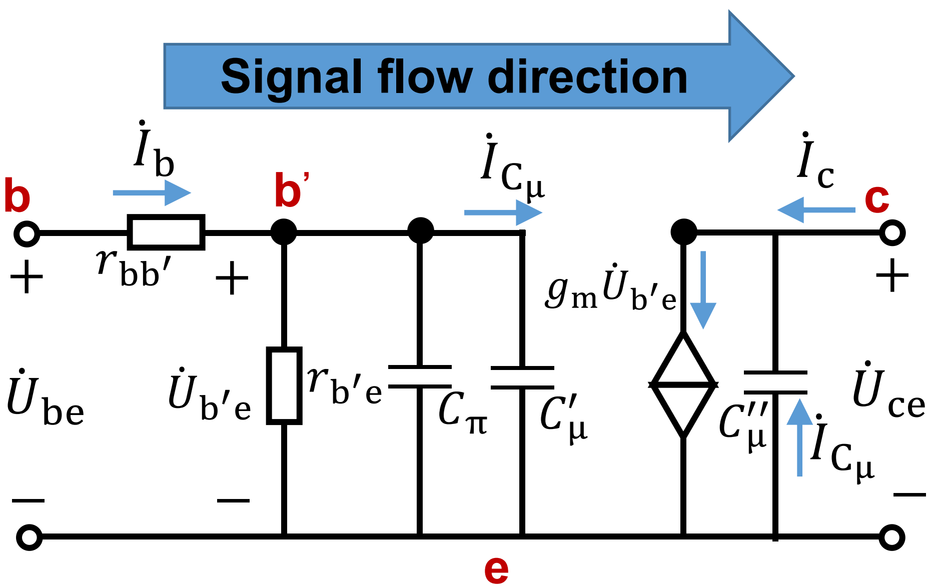
$$\begin{cases} \dot I_{C_\mu} = \dfrac{\dot U_{b'e} - \dot U_{ce}}{X_{C_\mu}}, \quad \dot K = \dfrac{\dot U_{ce}}{\dot U_{b'e}} \\ \dot I_{C_\mu} = \dfrac{\dot U_{b'e}}{X_{C'_\mu}} \\ \dot I_{C_\mu} = -\dfrac{\dot U_{ce}}{X_{C''_\mu}} \end{cases}$$

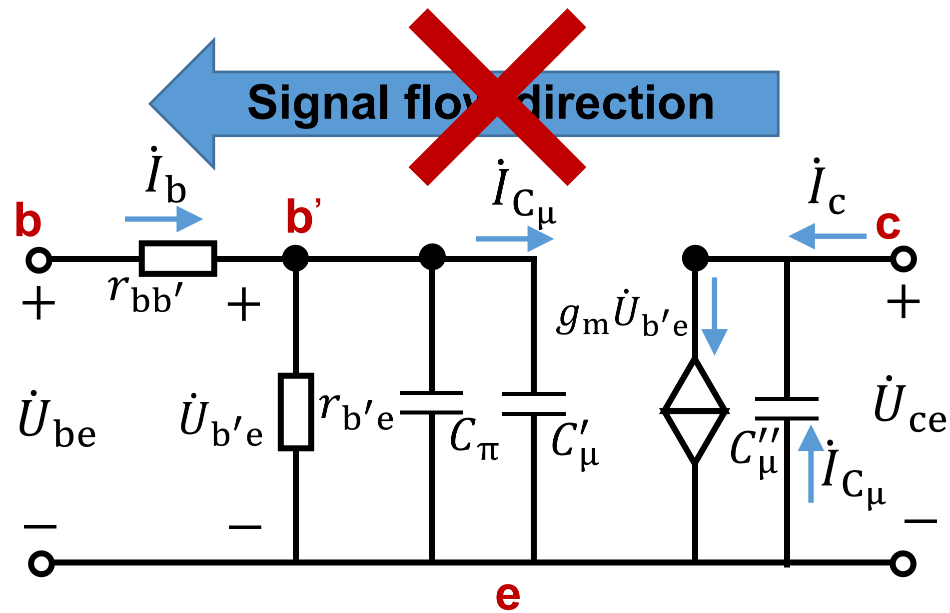
$$\begin{cases} X_{C'_\mu} = \dfrac{X_{C_\mu}}{(1-\dot K)} \\ X_{C''_\mu} = \dfrac{\dot K}{(\dot K - 1)} X_{C_\mu} \end{cases}$$

$$\begin{cases} C'_\mu = (1-\dot K) C_\mu \\ C''_\mu = \dfrac{\dot K - 1}{\dot K} C_\mu \end{cases}$$

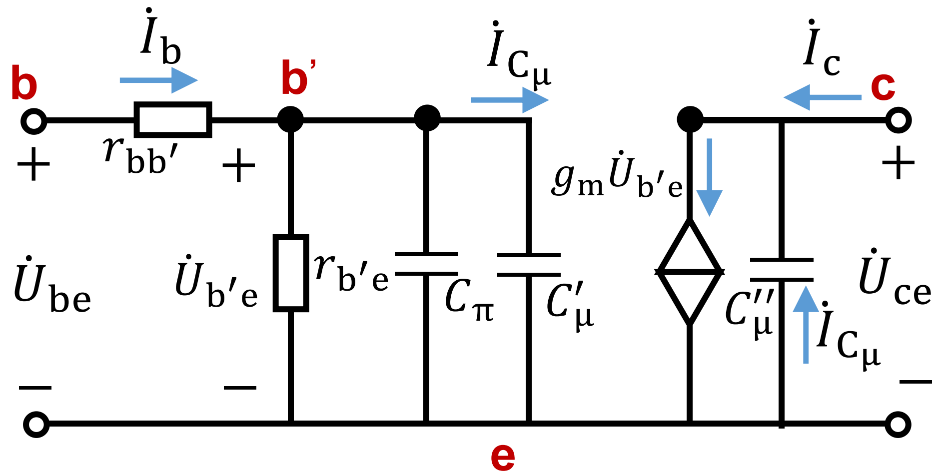
Unilateralization of hybrid - π model 混合π模型单向化

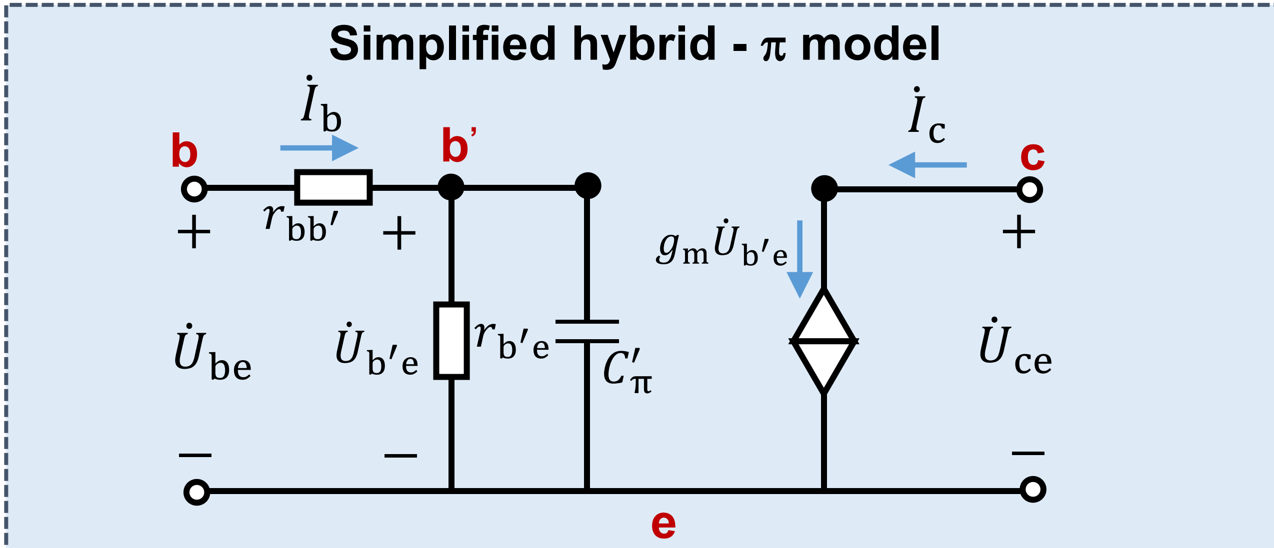






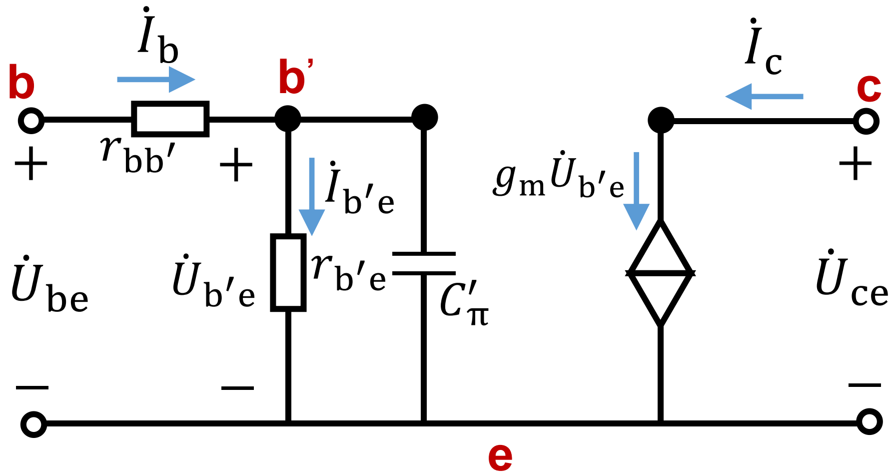
Simplified hybrid - π model 简化混合π模型





$$C'_\pi = C_\pi + C'_\mu = C_\pi + (1 - \dot{K})C_\mu$$

Simplified hybrid - π model



$$r_{b'e} = \beta_0 \frac{U_T}{|I_{CQ}|}$$ $, U_T = 26\text{mV}$

$$\begin{cases} \dot{I}_c = g_m \dot{U}_{b'e} = \beta_0 \dot{I}_{b'e} \\ \dot{U}_{b'e} = \dot{I}_{b'e} r_{b'e} \end{cases}$$

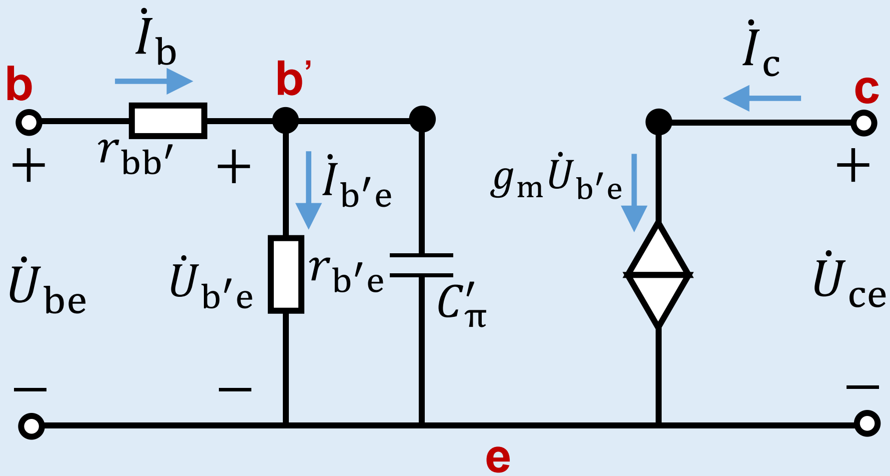
β_0: current gain at low frequency

$$C'_\pi = C_\pi + C'_\mu = C_\pi + (1 - \dot{K})C_\mu$$

$$g_m = \frac{\beta_0}{r_{b'e}} \approx \frac{I_{CQ}}{U_T}$$

$r_{bb'}, C_\mu$ are known   How to get C_π ?

Simplified hybrid - π model



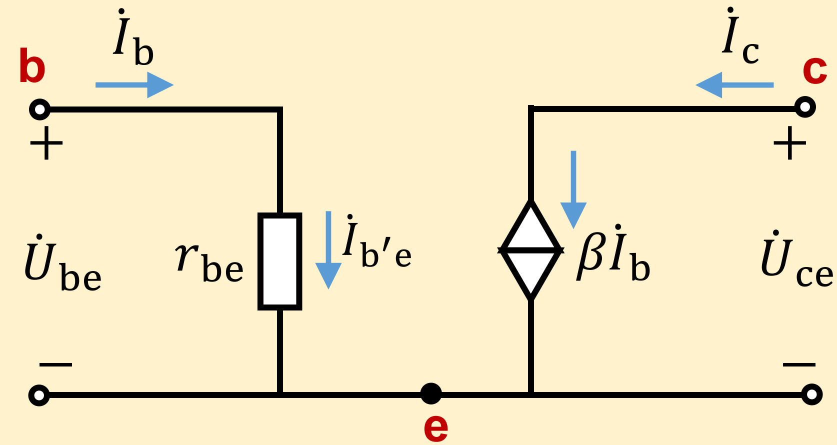
$$r_{\mathrm{b'e}} = \beta_0 \frac{U_{\mathrm{T}}}{|I_{\mathrm{CQ}}|} \quad , U_T = 26\mathrm{mV}$$

$$C'_{\pi} = C_{\pi} + (1 - \dot{K})\mathrm{C}_{\mu}$$

$$g_{\mathrm{m}} = \frac{\beta_0}{r_{\mathrm{b'e}}} \approx \frac{I_{\mathrm{CQ}}}{U_{\mathrm{T}}}$$

$$\dot{I}_{\mathrm{c}} = g_{\mathrm{m}}\dot{U}_{\mathrm{b'e}} = \beta_0 \dot{I}_{\mathrm{b'e}}$$

Simplified *h*-model



$$r_{\mathrm{be}} = r_{\mathrm{bb\prime}} + \beta_0 \frac{U_{\mathrm{T}}}{|I_{\mathrm{CQ}}|}$$

$$\beta_0 = \frac{\dot{I}_{\mathrm{c}}}{\dot{I}_{\mathrm{b}}}$$

$$\dot{I}_{\mathrm{c}} = \beta_0 \dot{I}_{\mathrm{b}}$$

5.3 Current amplification factor at high frequency

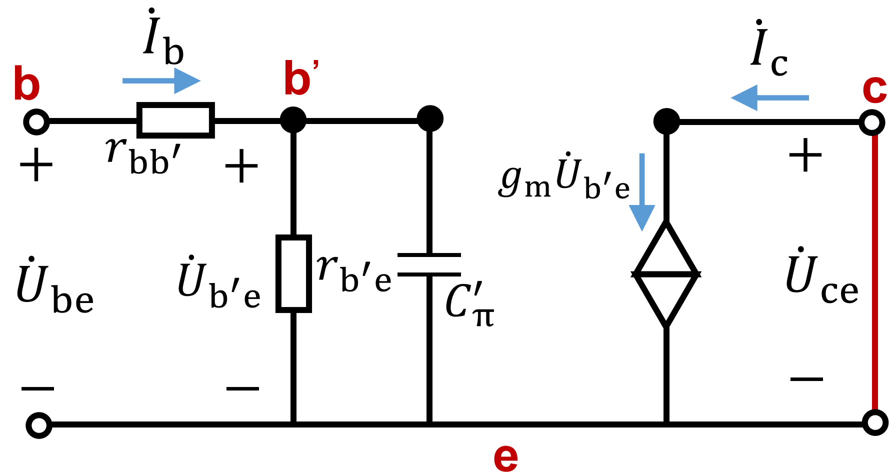
The current amplification factor is defined as

$$\dot{\beta} = \frac{\dot{I}_c}{\dot{I}_b}\bigg|_{U_{CE} = \text{ constant}}$$

$$\because \dot{K} = 0,$$

$$\therefore C'_\pi = C_\pi + \left(1 - \dot{K}\right)C_\mu = C_\pi + C_\mu$$

$$\begin{cases} \dot{I}_c = g_m \dot{U}_{b'e} = \dfrac{\beta_0}{r_{b'e}} \dot{U}_{b'e} \\ \dot{I}_b = \dfrac{\dot{U}_{b'e}}{r_{b'e}} + \dfrac{\dot{U}_{b'e}}{\dfrac{1}{j\omega C'_\pi}} \end{cases}$$



$$\dot{\beta} = \frac{\beta_0}{1 + j\omega r_{b'e} C'_\pi}$$

$$\boxed{\dot{\beta} = \frac{\beta_0}{1 + j\dfrac{f}{f_\beta}}}, f = \frac{\omega}{2\pi}, f_\beta = \frac{\omega_\beta}{2\pi} = \frac{1}{2\pi r_{b'e} C'_\pi}$$

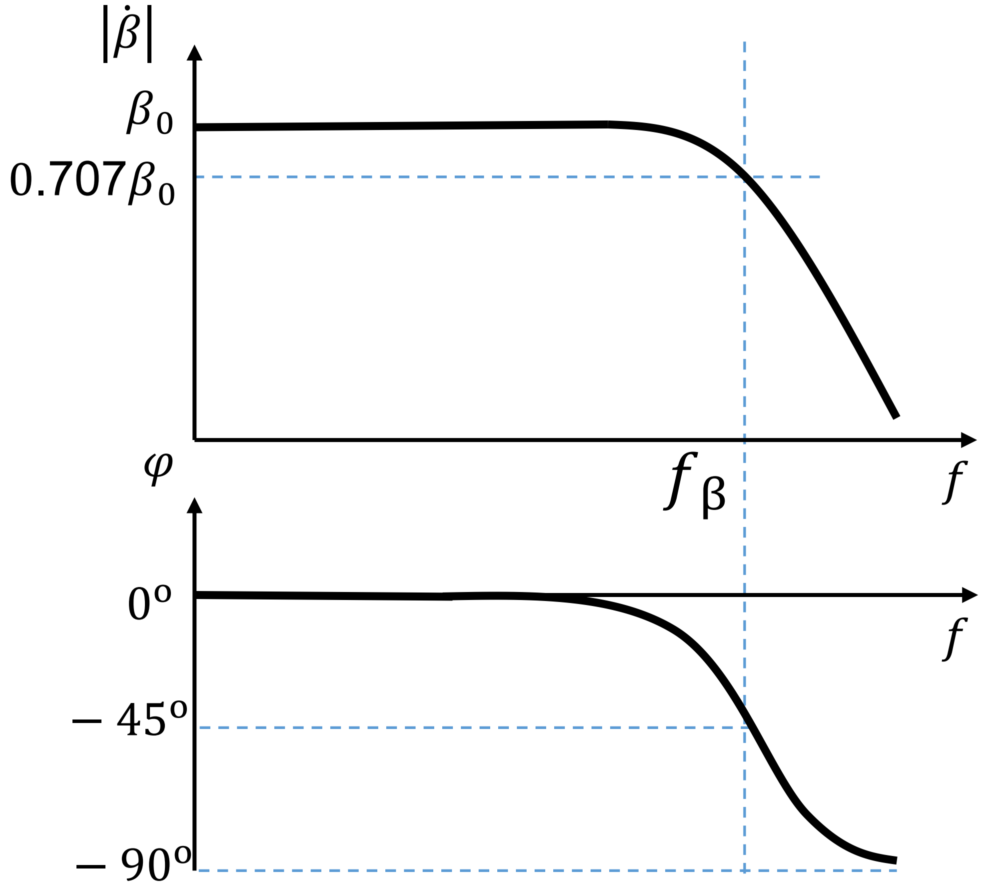
f_β: common-e cut-off frequency共射截止频率

The amplitude and phase frequency characteristics of $\dot{\beta}$

$$\begin{cases} \left|\dot{\beta}\right| = \dfrac{\beta_0}{\sqrt{1+\dfrac{f^2}{{f_\beta}^2}}} \\ \varphi = -\arctan\dfrac{f}{f_\beta} \end{cases}$$

$$f_\beta = \frac{1}{2\pi r_{b'e} C'_\pi} = \frac{1}{2\pi r_{b'e}(C_\pi + C_\mu)}$$

We can get C_π from f_β!



Bode plot of $\dot{\beta}$ 波特图

Use log scale for $|\dot{\beta}|$ and f 对数坐标

Use dB 分贝 1dB=20lg

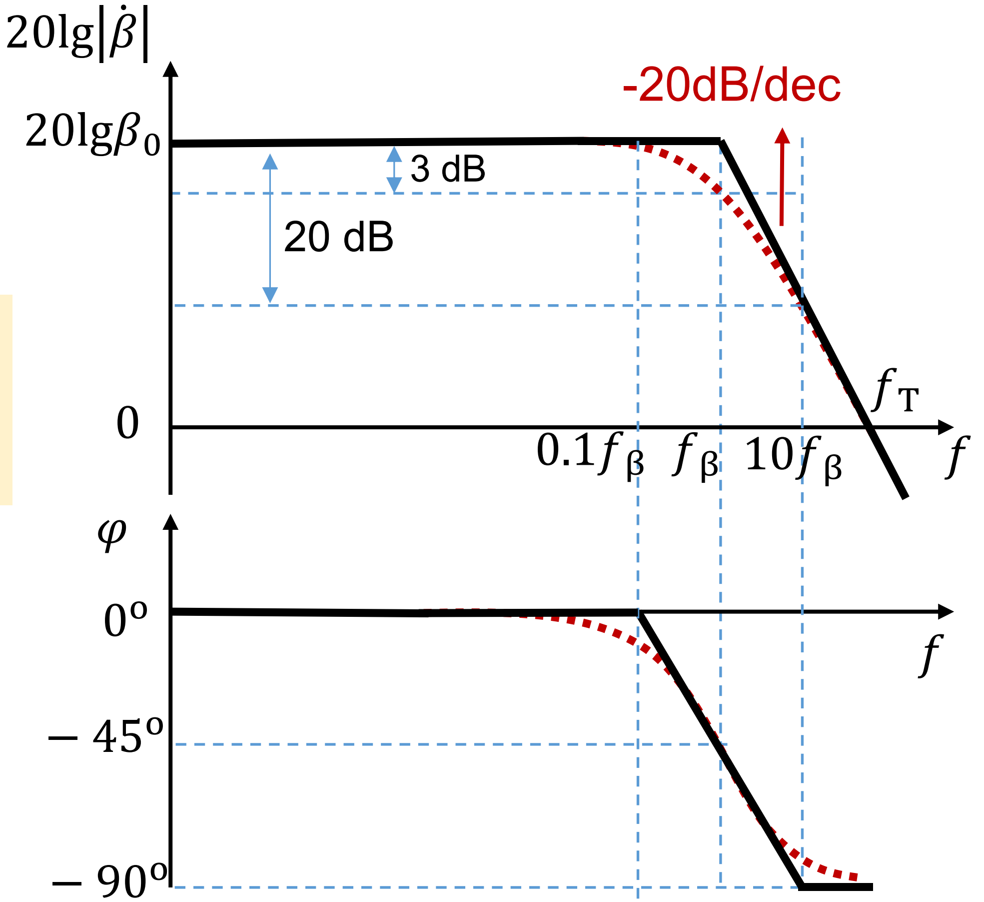
$$20\lg|\dot{\beta}| = 20\lg\beta_0 - 20\lg\sqrt{1+\frac{f^2}{{f_\beta}^2}}$$

$$20\lg\sqrt{2} \approx 3\ \text{dB}$$

$$\varphi = -\arctan\frac{f}{f_\beta}$$

$\because |\dot{\beta}| = 1$ or $20\lg|\dot{\beta}| = 0,\ f = f_T$

$\therefore f_T \approx \beta_0 f_\beta$



The common-base current amplification factor $\dot{\alpha}$

$$\boxed{\dot{\alpha}=\frac{\dot{\beta}}{1+\dot{\beta}}},\ \dot{\beta}=\frac{\beta_0}{1+j\dfrac{f}{f_\beta}} \qquad \dot{\alpha}=\frac{\dfrac{\beta_0}{1+\beta_0}}{1+j\dfrac{f}{(1+\beta_0)f_\beta}}$$

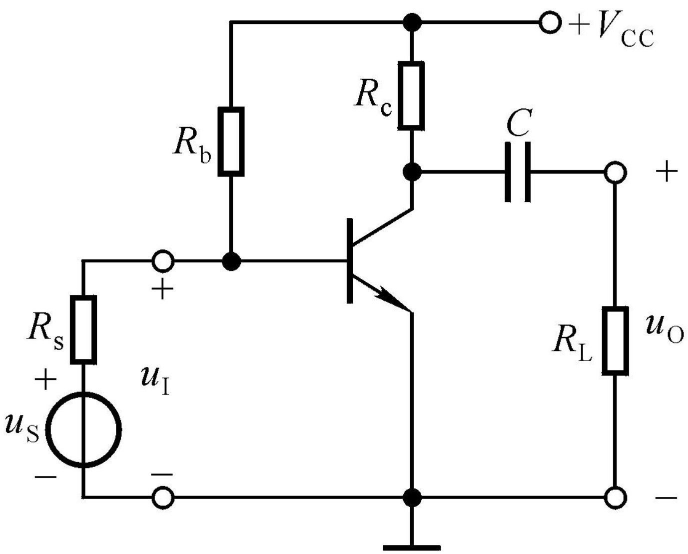
$$\boxed{\dot{\alpha}=\frac{\alpha_0}{1+j\dfrac{f}{f_\alpha}}}$$

Common-b cut-off frequency 共基极截止频率

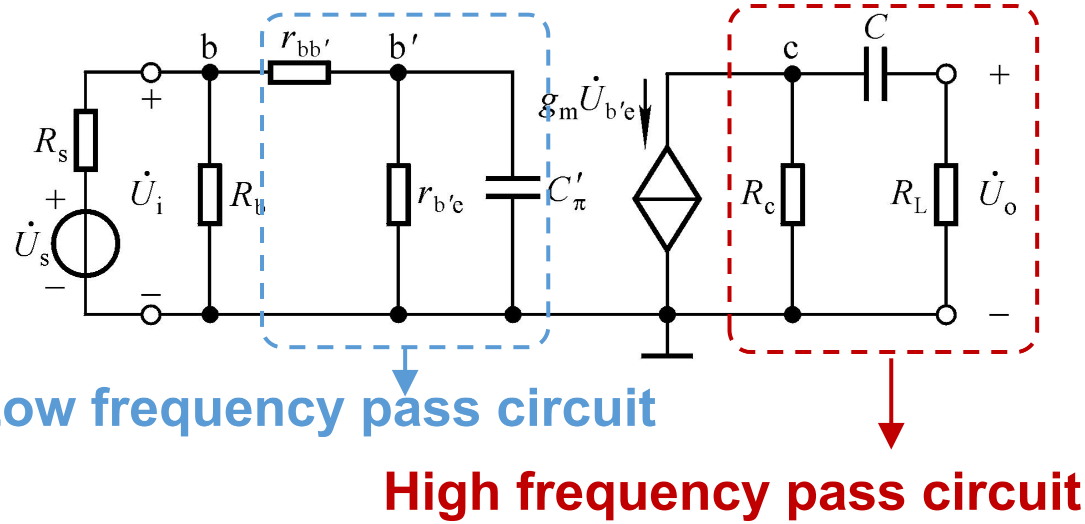
$$\boxed{f_\alpha=(1+\beta_0)f_\beta}$$

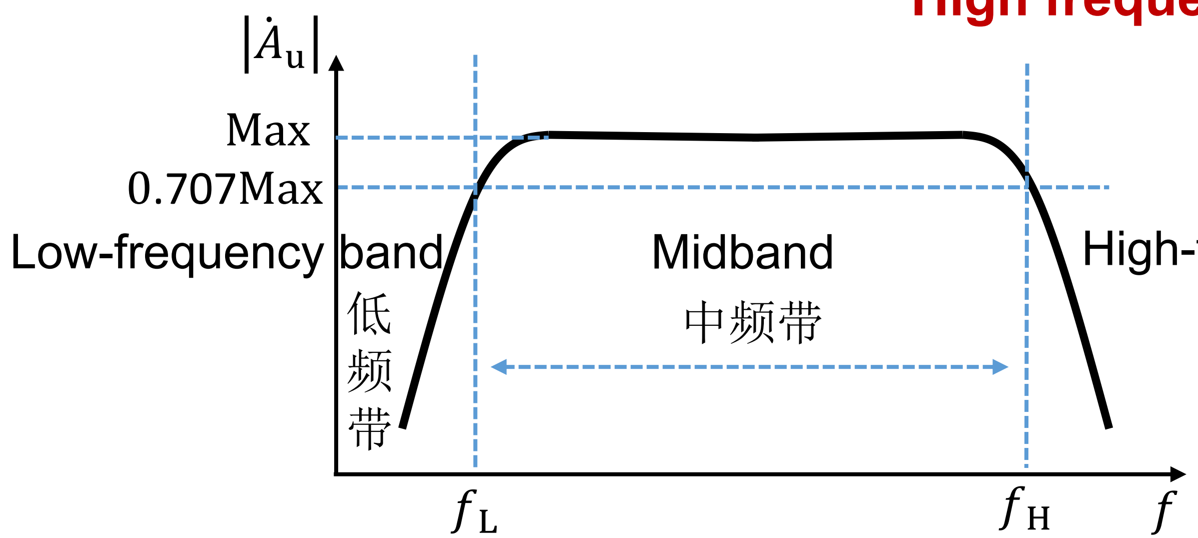
Common-b amplifier is more suitable for high frequency amplifier circuit

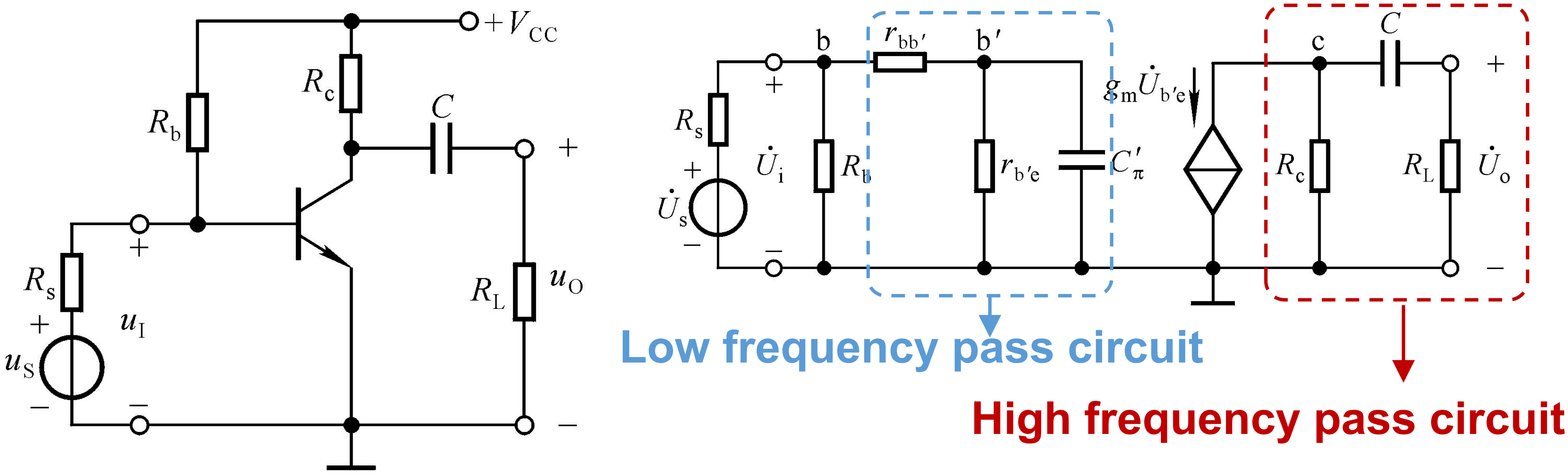
5.4 Frequency response of amplifier circuit



The AC circuit for: 0-∞ Hz







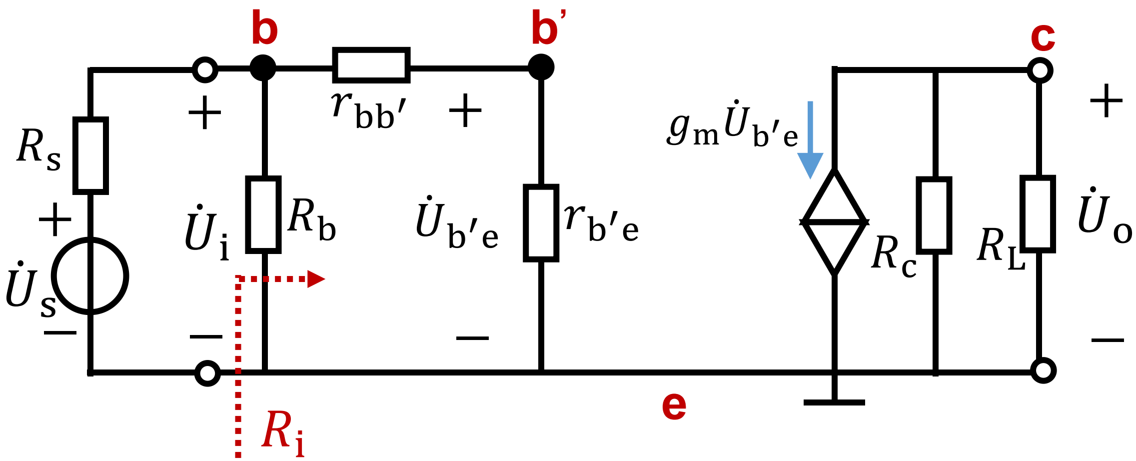
Analyze this circuit in three situations:

1) Low frequency: ignore C'_{π} (open), and consider C

2) Medium frequency: ignore C'_{π} (open), and C (short)

3) High frequency: ignore C (short), and consider C'_{π}

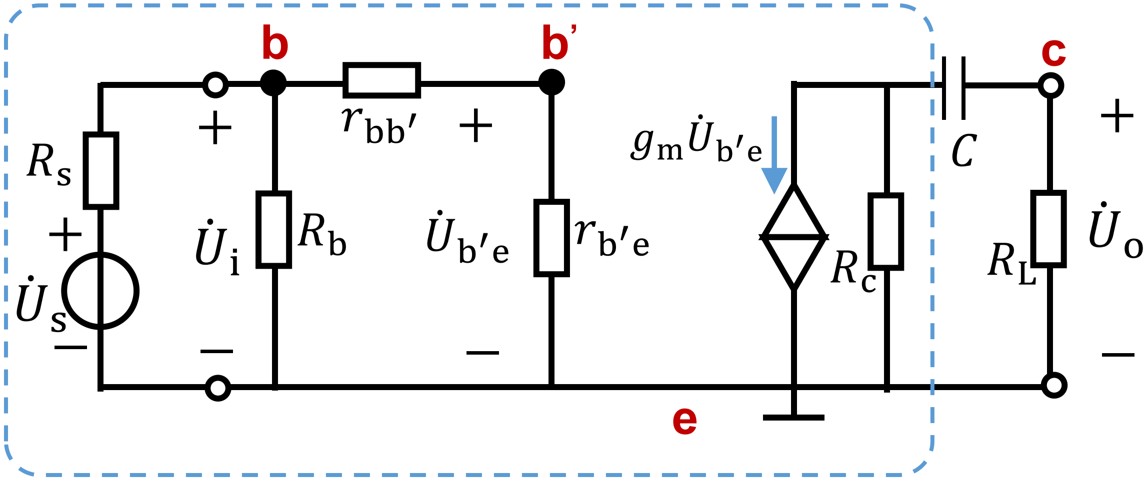
Medium frequency: ignore C'_π (open), and C (short)



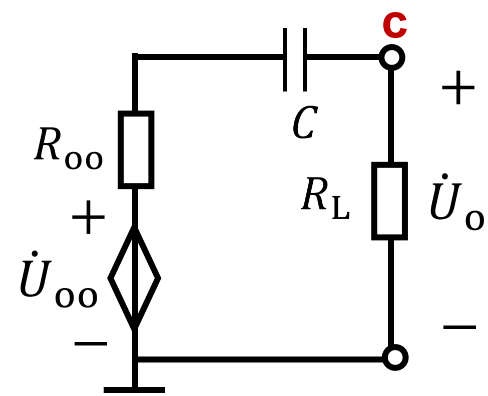
$$\dot{A}_{usm} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}_i}{\dot{U}_s} \cdot \frac{\dot{U}_{b'e}}{\dot{U}_i} \cdot \frac{\dot{U}_o}{\dot{U}_{b'e}} = \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot (-g_m R'_L)$$

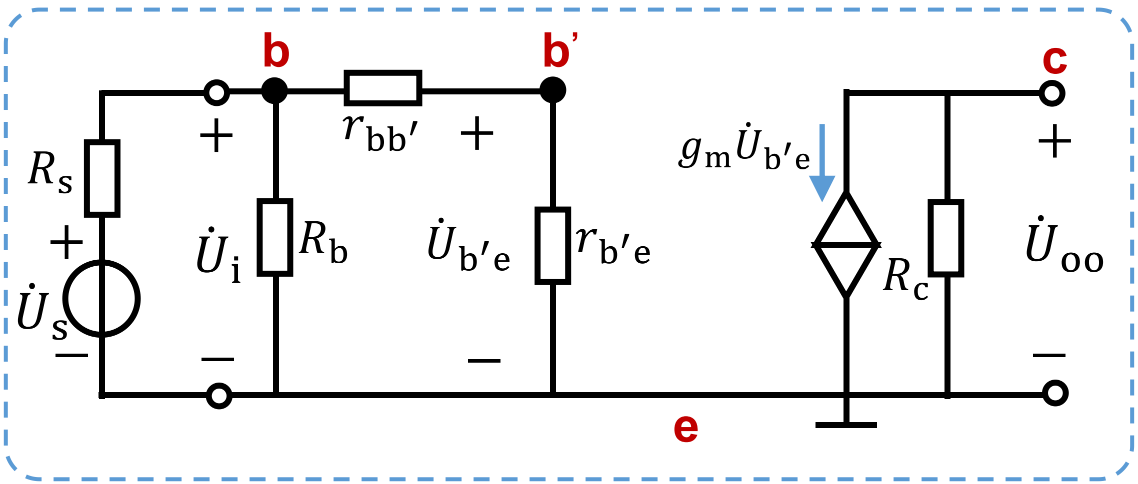
$$R_i = R_b \,//\, r_{be} \qquad R'_L = R_c \,//\, R_L$$

Low frequency: ignore C'_π (open), and consider C



Equivalent circuit

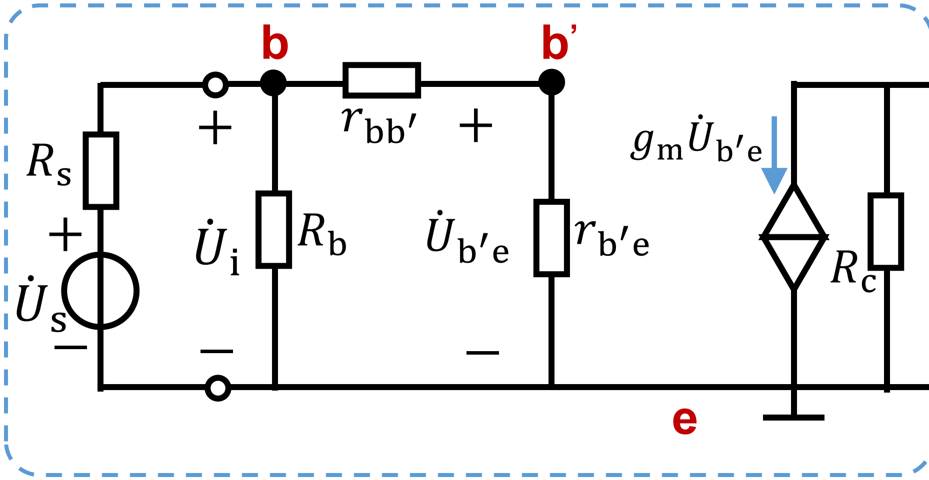


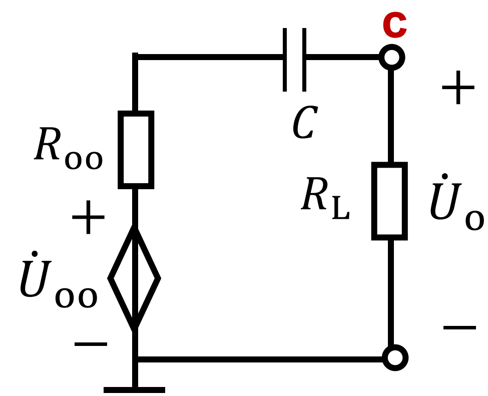


$$\dot{A}_{usl} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}_o}{\dot{U}_{oo}}\frac{\dot{U}_{oo}}{\dot{U}_s}$$

$$\frac{\dot{U}_{oo}}{\dot{U}_s} = \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot (-g_m R_c),\ R_{oo} = R_c$$

Low frequency: ignore C'_{π} (open), and consider C

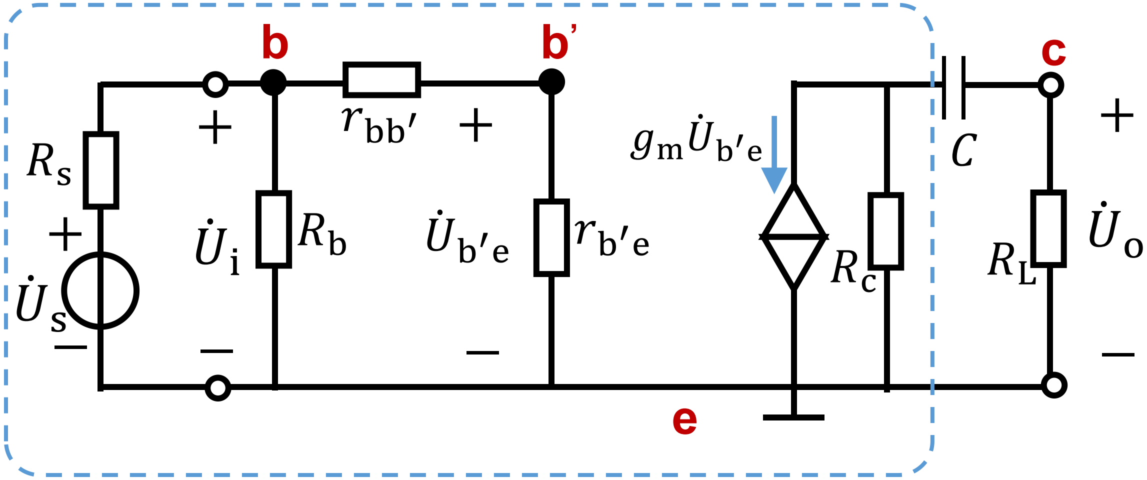


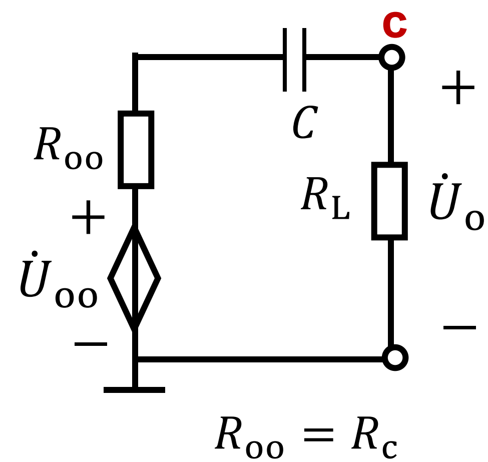


$$\dot{A}_{usl}=\frac{\dot{U}_o}{\dot{U}_s}=\frac{\dot{U}_{oo}}{\dot{U}_s}\frac{\dot{U}_o}{\dot{U}_{oo}}=\frac{R_i}{R_s+R_i}\cdot\frac{r_{b'e}}{r_{be}}\cdot(-\,g_m R_c)\frac{R_L}{R_c+R_L+\frac{1}{j\omega C}}$$

$$=\frac{R_i}{R_s+R_i}\cdot\frac{r_{b'e}}{r_{be}}\cdot\left(-\,g_m\frac{R_cR_L}{R_c+R_L}\right)\frac{1}{1+\frac{1}{j\omega(R_c+R_L)C}}$$

Low frequency: ignore C'_π (open), and consider C



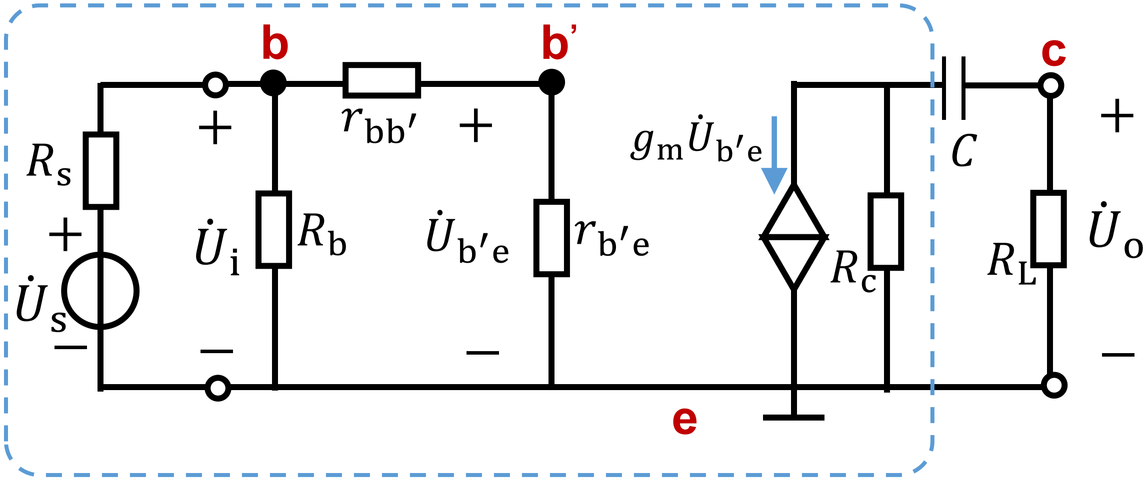


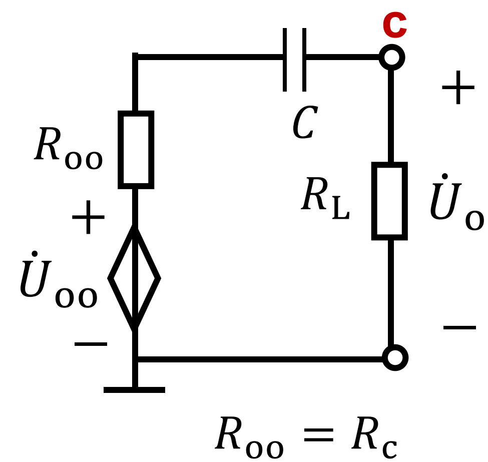
$$\dot{A}_{usl} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot \left(-g_m \frac{R_c R_L}{R_c + R_L}\right) \frac{1}{1 + \dfrac{1}{j\omega(R_c + R_L)C}}$$

$$= \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot (-g_m R'_L) \frac{j\,{}^{f}/_{f_L}}{1 + j\,{}^{f}/_{f_L}}$$

$$R'_L = \frac{R_c R_L}{R_c + R_L} \qquad f_L = \frac{1}{2\pi(R_c + R_L)C}$$

Low frequency: ignore C'_π (open), and consider C





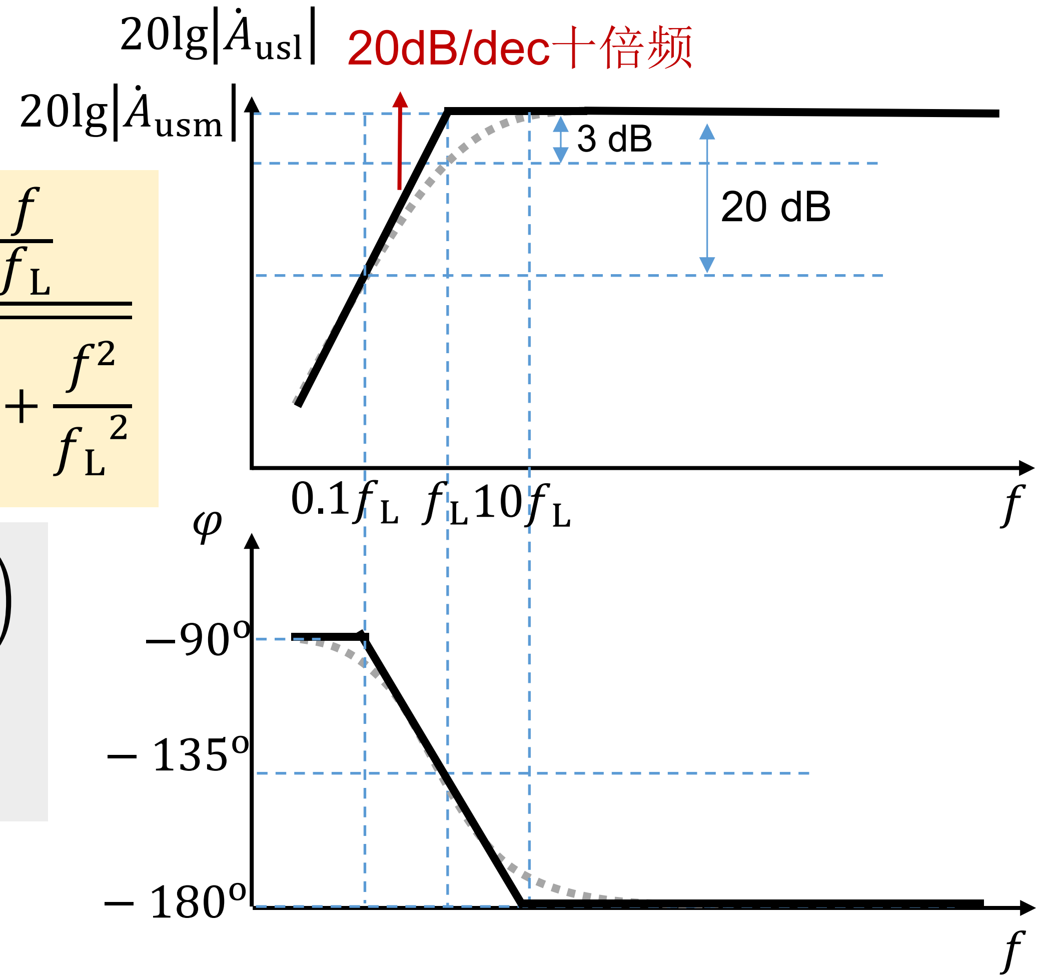
$$\dot{A}_{usl} = \frac{\dot{U}_o}{\dot{U}_s} = \underbrace{\frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot (-g_m R'_L)}_{\dot{A}_{usm}} \underbrace{\frac{j^{f}/_{f_L}}{1 + j^{f}/_{f_L}}}_{\text{High pass}}$$

$$\dot{A}_{usl} = \dot{A}_{usm} \frac{j^{f}/_{f_L}}{1 + j^{f}/_{f_L}}$$

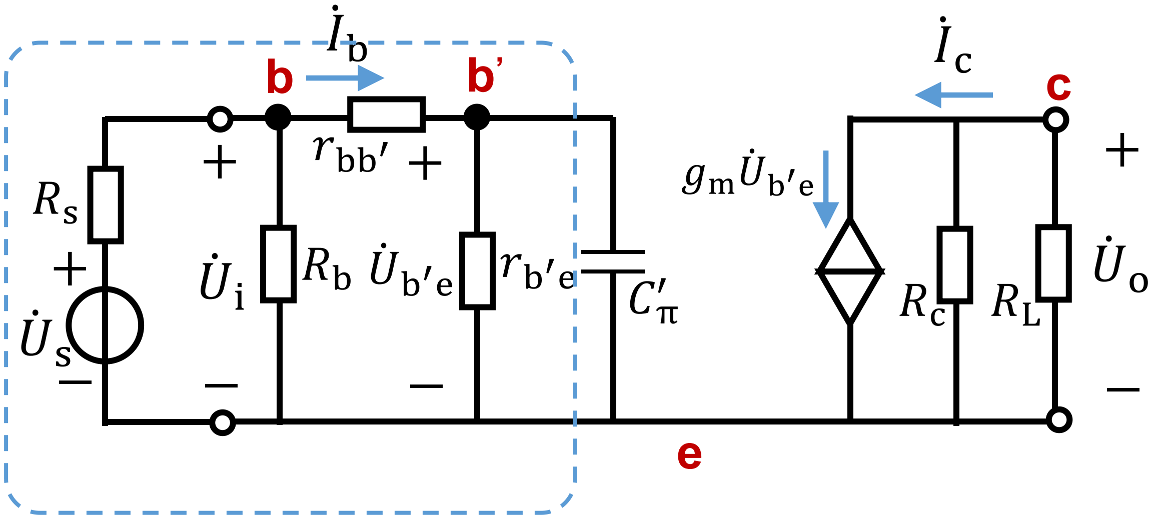
Low frequency: Amplitude and phase frequency characteristics

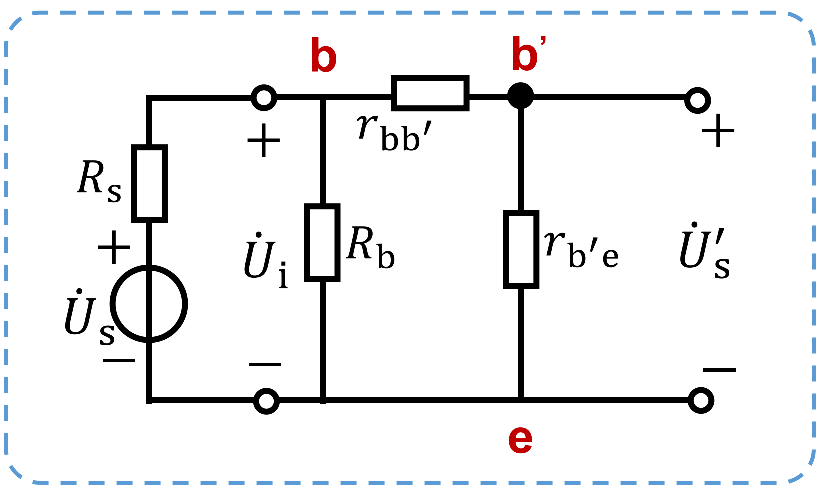
$$20\lg\left|\dot{A}_{usl}\right| = 20\lg\left|\dot{A}_{usm}\right| + 20\lg\frac{\frac{f}{f_L}}{\sqrt{1+\frac{f^2}{{f_L}^2}}}$$

$$\varphi = -180^o + \left(90^o - \arctan\frac{f}{f_L}\right)$$
$$= -90^o - \arctan\frac{f}{f_L}$$

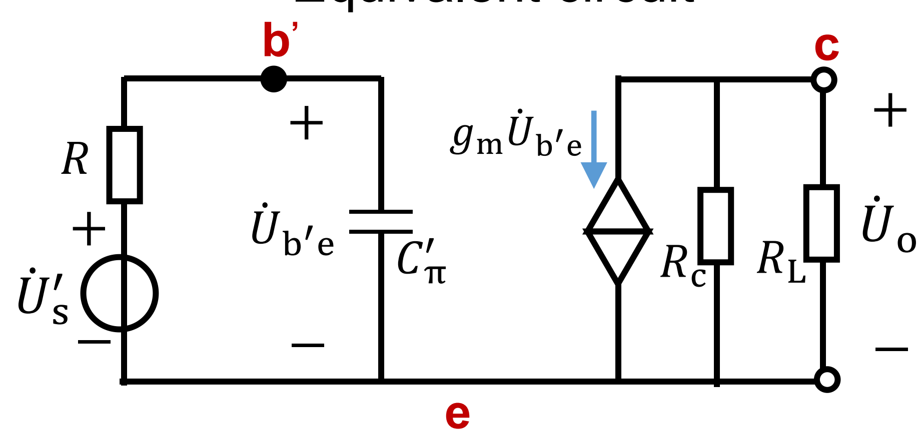


High frequency: ignore C (short) , and consider C'_π (open)

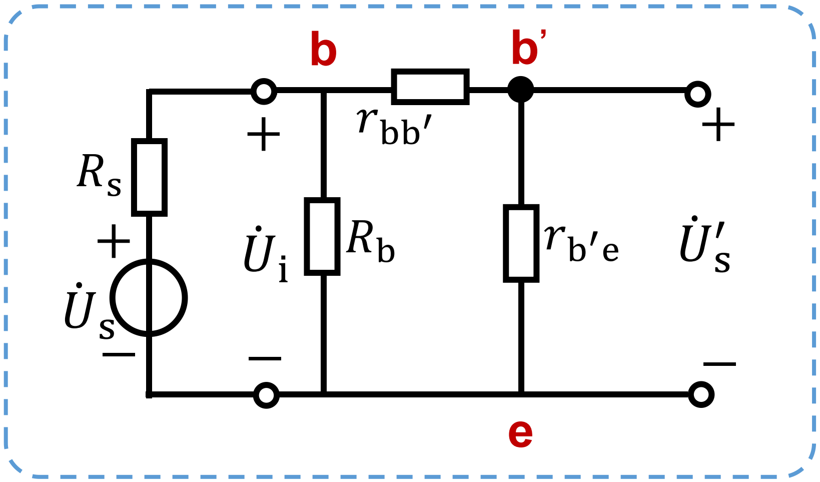


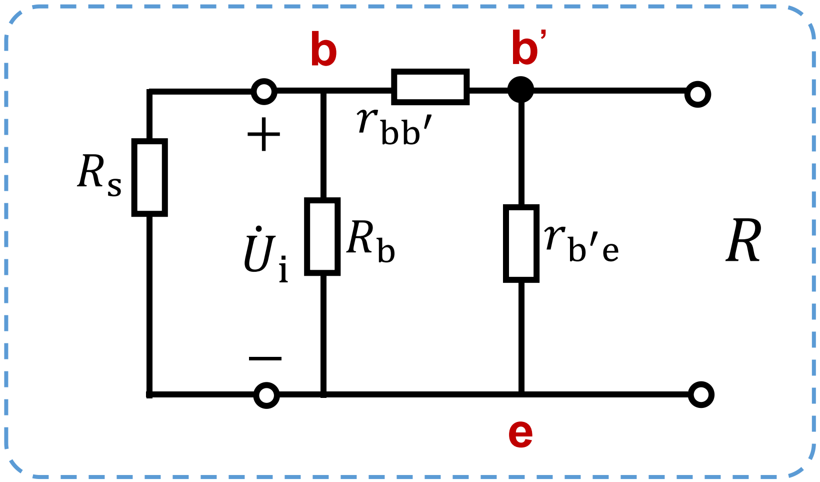


Equivalent circuit



High frequency: ignore C (short) , and consider C'_π (open)





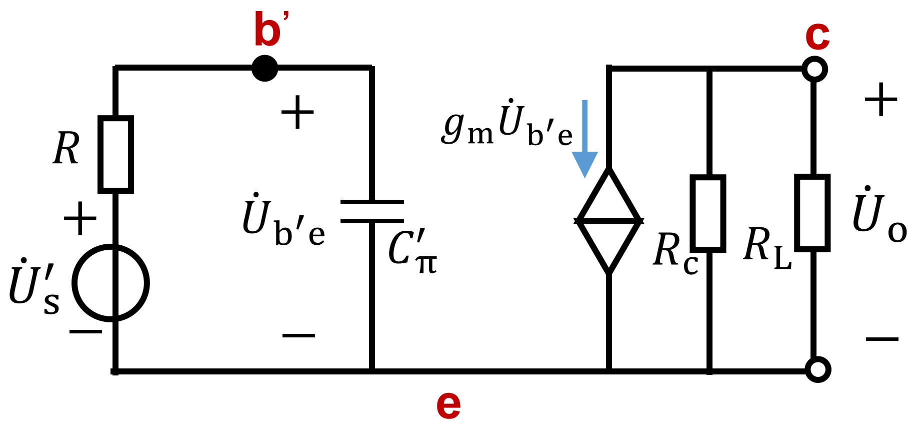
$$\dot{U}'_s = \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot \dot{U}_s$$

$$R_i = R_b \,//\, r_{be}$$

$$R = r_{b'e} \,//\, (r_{bb'} + R_s \,//\, R_b)$$

High frequency: ignore C (short) , and consider C'_π (open)

Equivalent circuit



$$\begin{cases} \dot{U}'_s = \dfrac{R_i}{R_s + R_i} \cdot \dfrac{r_{b'e}}{r_{be}} \cdot \dot{U}_s \\ R = r_{b'e} \,//\, (r_{bb'} + R_s \,//\, R_b) \end{cases}$$

$$\dot{A}_{ush} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}'_s}{\dot{U}_s} \cdot \frac{\dot{U}_o}{\dot{U}'_s} = \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot \frac{\dot{U}_o}{\dot{U}'_s}$$

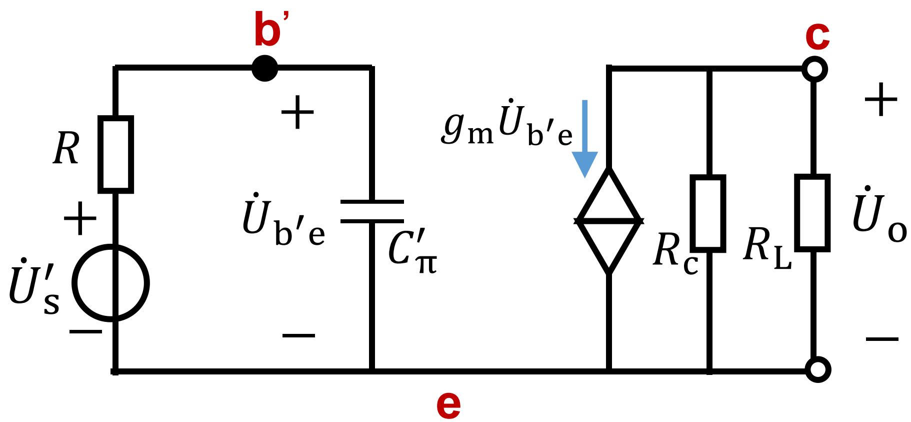
$$\begin{cases} \dot{U}_o = (-g_m \dot{U}_{b'e}) R'_L \\ \dot{U}_{b'e} = \dfrac{1/_{j\omega C'_\pi}}{R + 1/_{j\omega C'_\pi}} \dot{U}'_s \end{cases}$$

$$= \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot \frac{1/_{j\omega R C'_\pi}}{1 + 1/_{j\omega R C'_\pi}} \cdot (-g_m R'_L)$$

$$= \frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot (-g_m R'_L) \frac{1}{1 + j\, ^{f}/_{f_H}} \quad , f_H = \frac{1}{2\pi R C'_\pi} \qquad C'_\pi = C_\pi + (1 - \dot{K}) C_\mu$$

High frequency: ignore C (short) , and consider C'_π (open)

Equivalent circuit



$$C'_\pi = C_\pi + \left(1 - \dot{K}\right)C_\mu$$

$$\begin{cases} \dot{U}_o = \left(-g_m\dot{U}_{b'e}\right)R'_L \\ \dot{K} = \dfrac{\dot{U}_o}{\dot{U}_{b'e}} \end{cases}$$

$$f_H = \frac{1}{2\pi R C'_\pi}$$

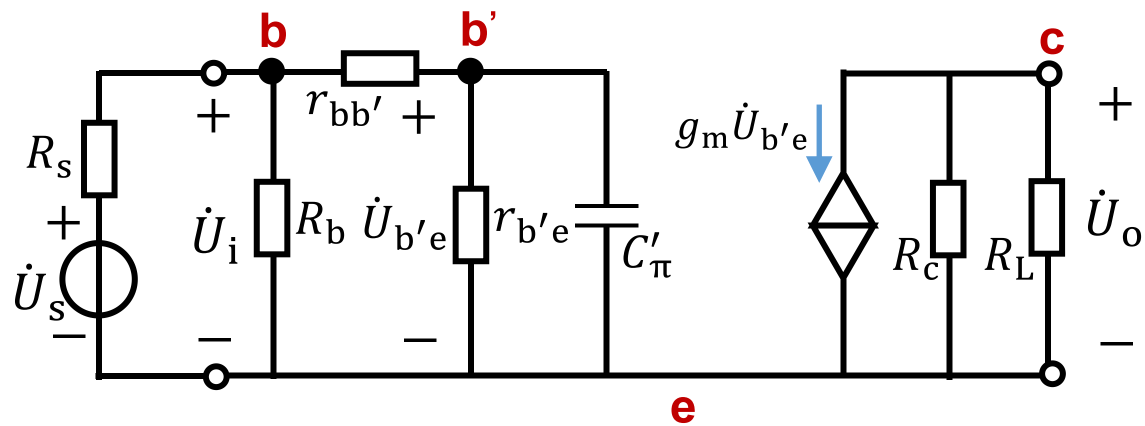
$$\dot{K} = -g_m R'_L$$

$$R = r_{b'e} \,//\, (r_{bb'} + R_s \,//\, R_b)$$

$$C'_\pi = C_\pi + (1 + g_m R'_L)C_\mu$$

C'_π ?

High frequency: ignore C (short) , and consider C'_π (open)



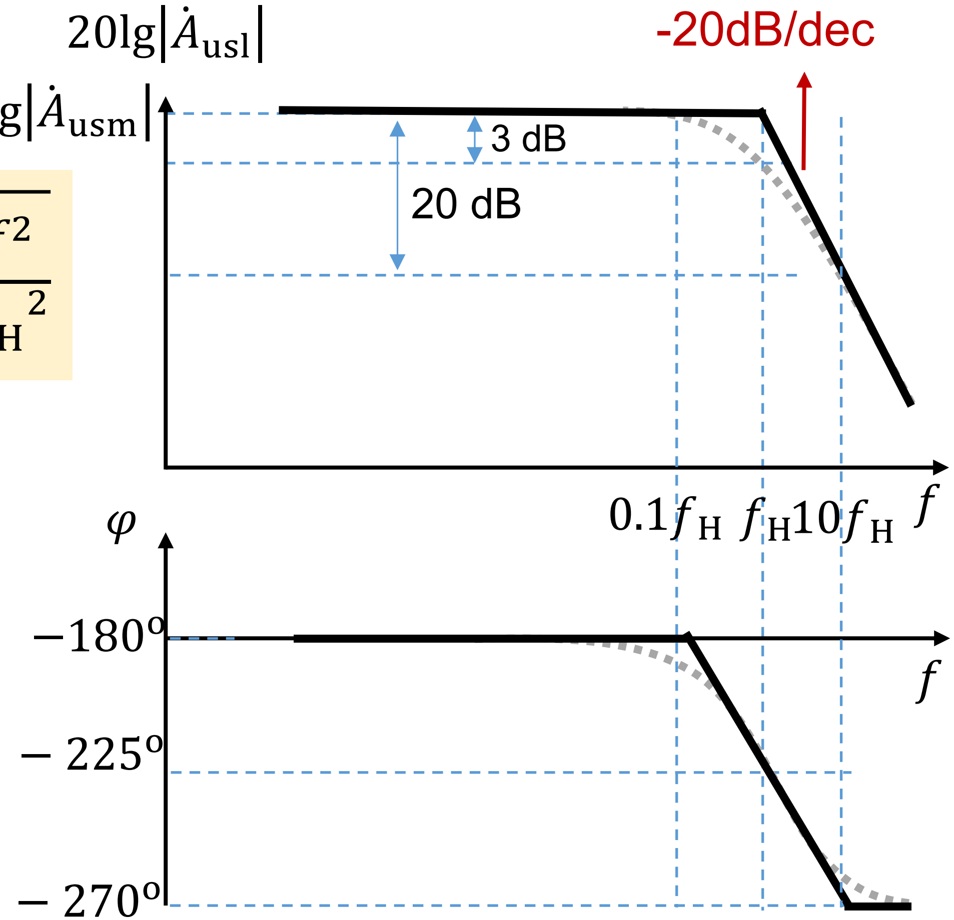
$$\dot A_{ush} = \frac{\dot U_o}{\dot U_s} = \underbrace{\frac{R_i}{R_s + R_i} \cdot \frac{r_{b'e}}{r_{be}} \cdot (-g_m R'_L)}_{\dot A_{usm}} \underbrace{\frac{1}{1 + j^{f}/_{f_H}}}_{\text{Low pass}}$$

$$\dot A_{ush} = \dot A_{usm} \frac{1}{1 + j^{f}/_{f_H}}$$

High frequency: Amplitude and phase frequency characteristics

$$20\lg\left|\dot{A}_{\mathrm{ush}}\right| = 20\lg\left|\dot{A}_{\mathrm{usm}}\right| - 20\lg\sqrt{1+\frac{f^2}{{f_{\mathrm{H}}}^2}}$$

$$\varphi = -180^{\mathrm{o}} - \arctan\frac{f}{f_{\mathrm{H}}}$$

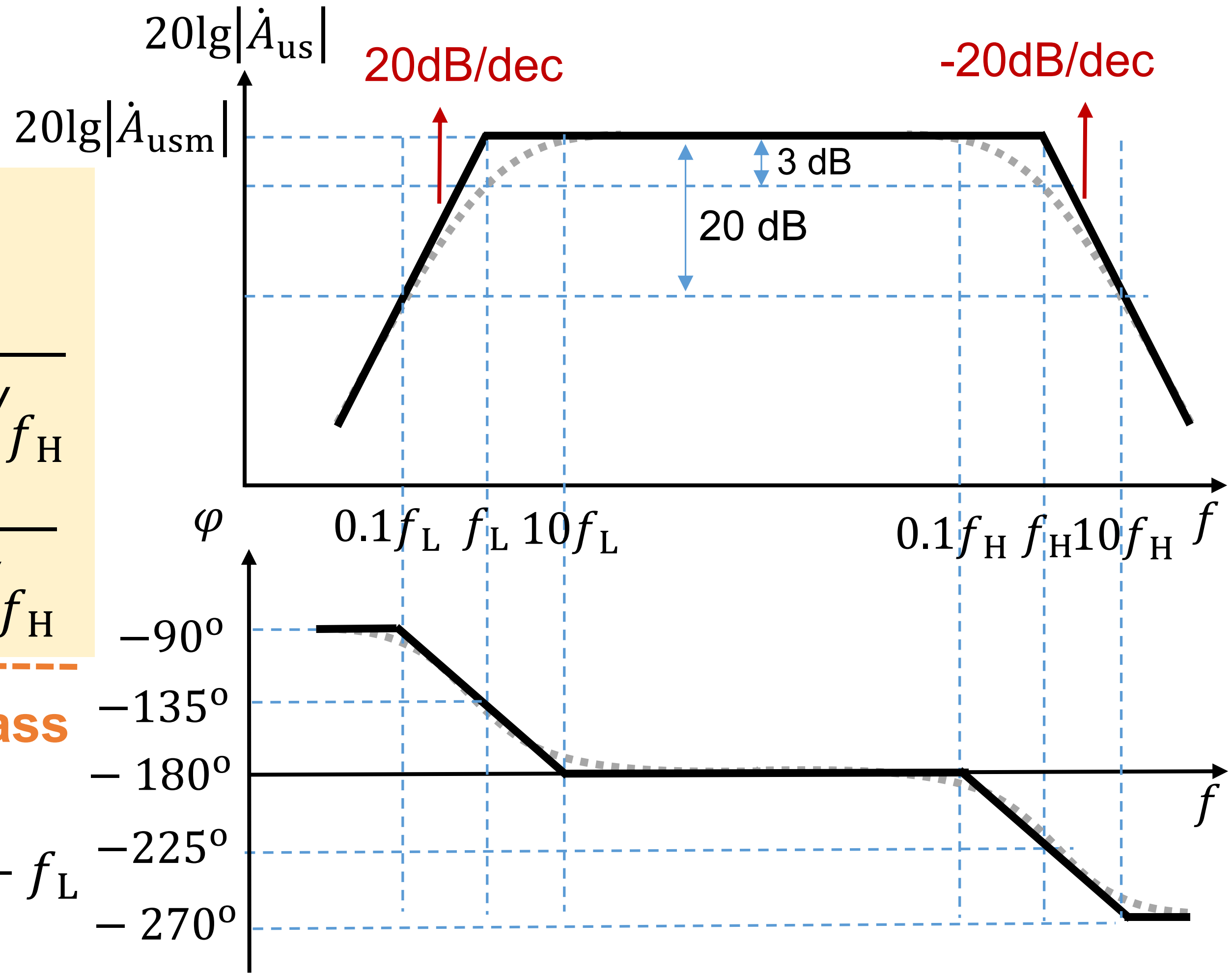


Full band: frequency response

$$\dot{A}_{us} = \dot{A}_{usm}\frac{j^{f}/_{f_L}}{1+j^{f}/_{f_L}}\frac{1}{1+j^{f}/_{f_H}} = \dot{A}_{usm}\frac{1}{1+{}^{f_L}/_{jf}}\frac{1}{1+j^{f}/_{f_H}}$$

High pass Low pass

Bandwidth $f_{bw} = f_H - f_L$



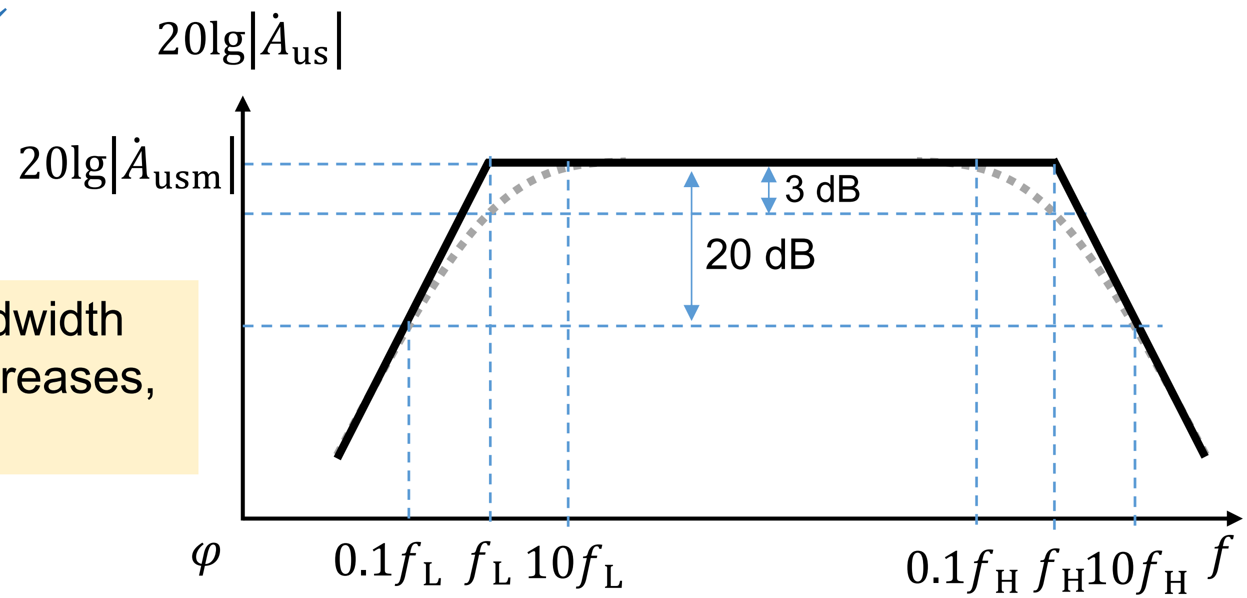
Bandwidth gain product: 带宽增益积

Bandwidth $f_{\mathrm{bw}} = f_{\mathrm{H}} - f_{\mathrm{L}} \approx f_{\mathrm{H}}$ $\begin{cases} f_{\mathrm{H}} = \dfrac{1}{2\pi[r_{\mathrm{b'e}} \,//\, (r_{\mathrm{bb'}} + R_{\mathrm{s}} \,//\, R_{\mathrm{b}})\,]C'_{\pi}} \\ C'_{\pi} = C_{\pi} + \ (1 + g_{\mathrm{m}} R'_{\mathrm{L}}) \mathrm{C}_{\mu} \end{cases}$

Voltage gain $\dot{A}_{\mathrm{usm}} = \dfrac{R_{\mathrm{i}}}{R_{\mathrm{s}} + R_{\mathrm{i}}} \cdot \dfrac{r_{\mathrm{b'e}}}{r_{\mathrm{be}}} \cdot (-g_{\mathrm{m}} R'_{\mathrm{L}})$

$$\begin{cases} g_{\mathrm{m}} R'_{\mathrm{L}} \uparrow \ \rightarrow C'_{\pi} \uparrow \rightarrow f_{\mathrm{H}} \downarrow \\ g_{\mathrm{m}} R'_{\mathrm{L}} \uparrow \ \rightarrow |\dot{A}_{\mathrm{usm}}| \uparrow \end{cases}$$

When gain increases, bandwidth decreases; When gain decreases, bandwidth increases.



Bandwidth gain product: 带宽增益积

$$|\dot{A}_{\mathrm{usm}} f_{\mathrm{bw}}| \approx |\dot{A}_{\mathrm{usm}} f_{\mathrm{H}}| = \frac{R_{\mathrm{i}}}{R_{\mathrm{s}} + R_{\mathrm{i}}} \cdot \frac{r_{\mathrm{b'e}}}{r_{\mathrm{be}}} \cdot \frac{g_{\mathrm{m}} R'_{\mathrm{L}}}{2\pi [r_{\mathrm{b'e}} // (r_{\mathrm{bb'}} + R_{\mathrm{s}} // R_{\mathrm{b}})] C'_{\pi}}$$

$$C'_{\pi} = C_{\pi} + (1 + g_{\mathrm{m}} R'_{\mathrm{L}}) \mathrm{C}_{\mu}, \quad R_{\mathrm{i}} = R_{\mathrm{b}} // r_{\mathrm{be}}, \quad R'_{\mathrm{L}} = R_{\mathrm{c}} // R_{\mathrm{L}}$$

If $r_{\mathrm{be}} \ll R_{\mathrm{b}}$, $R_{\mathrm{s}} \ll R_{\mathrm{b}}$, $g_{\mathrm{m}} R'_{\mathrm{L}} \gg 1$, $g_{\mathrm{m}} R'_{\mathrm{L}} \mathrm{C}_{\mu} \gg \mathrm{C}_{\pi}$, we have:

$$R_{\mathrm{i}} \approx r_{\mathrm{be}}, \quad C'_{\pi} \approx g_{\mathrm{m}} R'_{\mathrm{L}} \mathrm{C}_{\mu},$$

$$\boxed{|\dot{A}_{\mathrm{usm}} f_{\mathrm{bw}}| \approx \frac{1}{2\pi (r_{\mathrm{bb'}} + R_{\mathrm{s}}) \mathrm{C}_{\mu}}}$$

Bandwidth gain product: 带宽增益积

$$|\dot{A}_{\mathrm{usm}} f_{\mathrm{bw}}| \approx \frac{1}{2\pi\,(r_{\mathrm{bb'}} + R_{\mathrm{s}})\mathrm{C}_{\mu}}$$

$|\dot{A}_{\mathrm{usm}} f_{\mathrm{bw}}|$ only depends on the property of amplifier:$r_{\mathrm{bb'}}$ and C_{μ}.

When amplifier is fixed, the bandwidth gain product of the amplifier circuit is fixed.

To improve the high frequency performance:

- Use amplifier with smaller $r_{\mathrm{bb'}}$ and C_{μ} values.
- Use common-base amplifier circuit

5.5 Frequency response of multistage amplifier circuit

A simple case: two-stage amplifier circuit

The first stage voltage gain: $\dot{A}_{\mathrm{u1}}$, second stage: $\dot{A}_{\mathrm{u2}} = \dot{A}_{\mathrm{u1}}$.

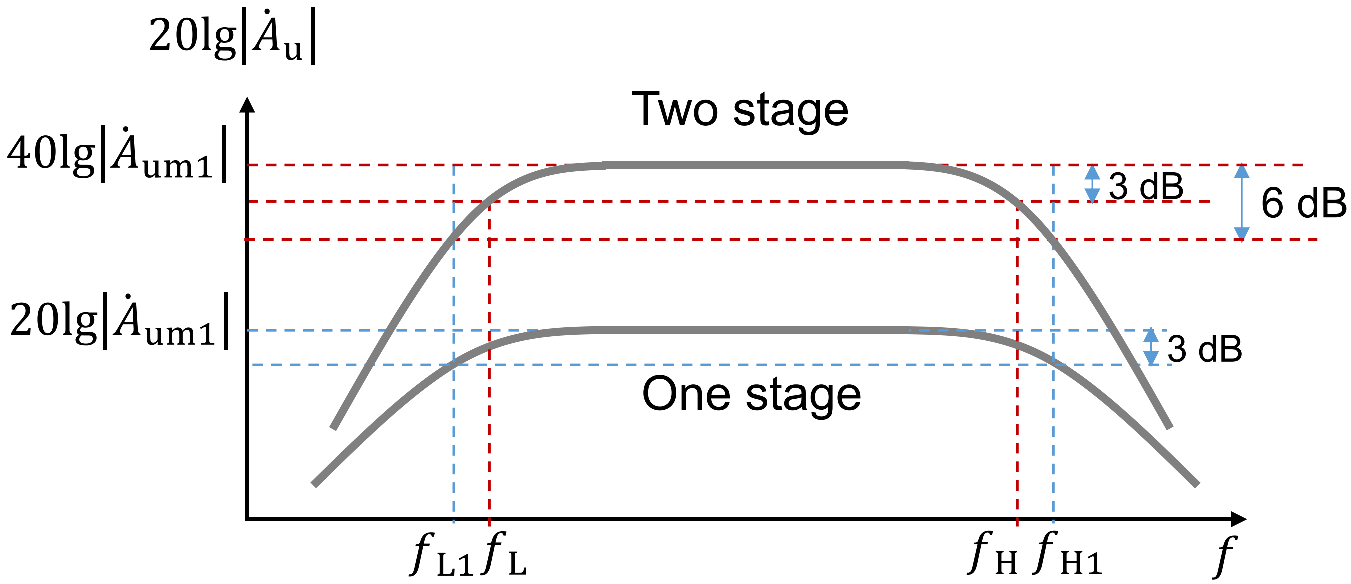
Two-stage voltage gain: $\dot{A}_{\mathrm{u}} = \dot{A}_{\mathrm{u1}} \cdot \dot{A}_{\mathrm{u2}}$

$$|\dot{A}_{\mathrm{u}}|e^{j\varphi} = |\dot{A}_{\mathrm{u1}}|e^{j\varphi_1} \cdot |\dot{A}_{\mathrm{u2}}|e^{j\varphi_2} = |\dot{A}_{\mathrm{u1}}| \cdot |\dot{A}_{\mathrm{u2}}|e^{j(\varphi_1+\varphi_2)}$$

$$|\dot{A}_{\mathrm{u}}| = |\dot{A}_{\mathrm{u1}}| \cdot |\dot{A}_{\mathrm{u2}}|$$

$$\begin{cases} \text{Amplitude} \quad 20\lg|\dot{A}_{\mathrm{u}}| = 20\lg|\dot{A}_{\mathrm{u1}}| + 20\lg|\dot{A}_{\mathrm{u2}}| = 40\lg|\dot{A}_{\mathrm{u1}}| \\ \text{Phase} \quad \varphi = \varphi_1 + \varphi_2 \end{cases}$$

Amplitude $20\lg|\dot{A}_u| = 20\lg|\dot{A}_{u1}| + 20\lg|\dot{A}_{u2}| = 40\lg|\dot{A}_{u1}|$



When $f = f_{L1}$ or $f = f_{H1}$, $20\lg|\dot{A}_{u1}| = 20\lg|\dot{A}_{um1}| - 3\text{dB}$

$$20\lg|\dot{A}_u| = 40\lg|\dot{A}_{u1}| = 40\lg|\dot{A}_{um1}| - 6\text{dB}$$

The bandwidth of two-stage amplifier circuit becomes narrower!

Q: f_L? f_H?

How to get the relation between f_{L1} (f_{H1}) and f_{L} (f_{H}) ?

One stage:

$$\begin{cases} 20\lg\left|\dot{A}_{\mathrm{uh1}}\right| = 20\lg\left|\dot{A}_{\mathrm{um1}}\right| - 20\lg\sqrt{1+\dfrac{f^2}{{f_{\mathrm{H1}}}^2}} & \textbf{High-f region} \\ 20\lg\left|\dot{A}_{\mathrm{ul1}}\right| = 20\lg\left|\dot{A}_{\mathrm{um1}}\right| + 20\lg\dfrac{\dfrac{f}{f_{\mathrm{L1}}}}{\sqrt{1+\dfrac{f^2}{{f_{\mathrm{L1}}}^2}}} & \textbf{Low-f region} \end{cases}$$

Two stage: $20\lg\left|\dot{A}_{\mathrm{u}}\right| = 40\lg\left|\dot{A}_{\mathrm{u1}}\right|$

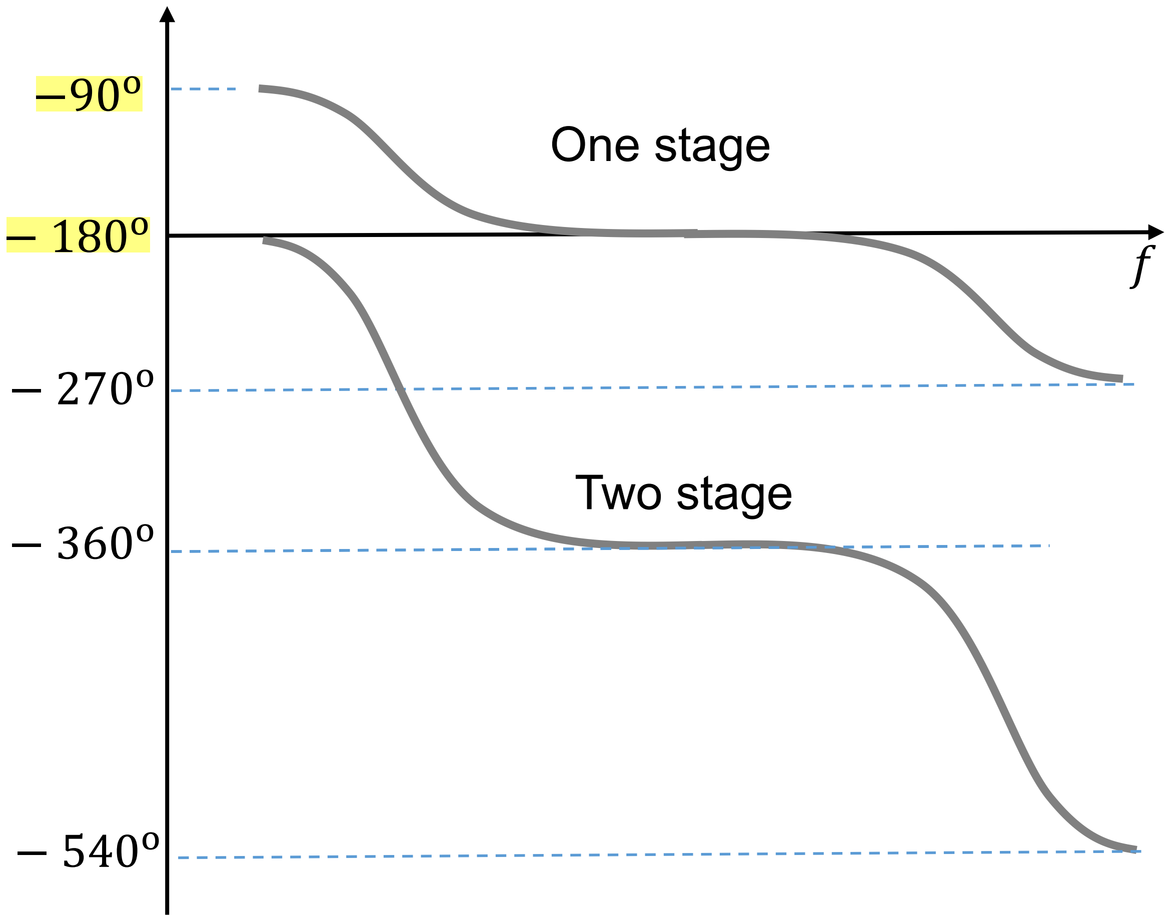
$$\begin{cases} 20\lg\left|\dot{A}_{\mathrm{uh}}\right| = 40\lg\left|\dot{A}_{\mathrm{um1}}\right| - 40\lg\sqrt{1+\dfrac{f^2}{{f_{\mathrm{H1}}}^2}} & \textbf{High-f region} \\ 20\lg\left|\dot{A}_{\mathrm{ul}}\right| = 40\lg\left|\dot{A}_{\mathrm{um1}}\right| + 40\lg\dfrac{\dfrac{f}{f_{\mathrm{L1}}}}{\sqrt{1+\dfrac{f^2}{{f_{\mathrm{L1}}}^2}}} & \textbf{Low-f region} \end{cases}$$

How to get the relation between f_{L1} (f_{H1}) and f_{L} (f_{H}) ?

When $f = f_{\mathrm{L}}$ or $f = f_{\mathrm{H}}$,

$$
\begin{cases}
40\lg\sqrt{1+\dfrac{{f_{\mathrm{H}}}^2}{{f_{\mathrm{H1}}}^2}} = 3\ \mathrm{dB} = 20\lg\sqrt{2} \\[2ex]
40\lg\dfrac{\dfrac{f_{\mathrm{L}}}{f_{\mathrm{L1}}}}{\sqrt{1+\dfrac{{f_{\mathrm{L}}}^2}{{f_{\mathrm{L1}}}^2}}} =-\ 3\ \mathrm{dB} =-\ 20\lg\sqrt{2}
\end{cases}
$$

Phase: $\varphi = \varphi_1 + \varphi_2$



N-stage amplifier circuit

N-stage voltage gain: $\dot{A}_{\mathrm{u}} = \dot{A}_{\mathrm{u1}} \cdot \dot{A}_{\mathrm{u2}} \cdots \dot{A}_{\mathrm{uN}}$

$$|\dot{A}_{\mathrm{u}}|e^{j\varphi} = |\dot{A}_{\mathrm{u1}}|e^{j\varphi_1} \cdot |\dot{A}_{\mathrm{u2}}|e^{j\varphi_2} \cdots |\dot{A}_{\mathrm{uN}}|e^{j\varphi_{\mathrm{N}}} = |\dot{A}_{\mathrm{u1}}| \cdot |\dot{A}_{\mathrm{u2}}| \cdots |\dot{A}_{\mathrm{uN}}|e^{j(\varphi_1+\varphi_2+\cdots+\varphi_{\mathrm{N}})}$$

$$\begin{cases} \text{Amplitude} & 20\lg|\dot{A}_{\mathrm{u}}| = 20\lg|\dot{A}_{\mathrm{u1}}| + 20\lg|\dot{A}_{\mathrm{u2}}| + \cdots + 20\lg|\dot{A}_{\mathrm{uN}}| \\ \text{Phase} & \varphi = \varphi_1 + \varphi_2 + \cdots + \varphi_{\mathrm{N}} \end{cases}$$

$$\begin{cases} \text{Amplitude} & 20\lg|\dot{A}_{\mathrm{u}}| = \sum_{\mathrm{k}=1}^{\mathrm{N}} 20\lg|\dot{A}_{\mathrm{uk}}| \\ \text{Phase} & \varphi = \sum_{\mathrm{k}=1}^{\mathrm{N}} \varphi_{\mathrm{k}} \end{cases}$$

N-stage amplifier circuit

$$\begin{cases} \text{Amplitude} \quad 20\lg|\dot{A}_{\mathrm{u}}| = \sum_{\mathrm{k=1}}^{\mathrm{N}} 20\lg|\dot{A}_{\mathrm{uk}}| \\ \text{Phase} \quad \varphi = \sum_{\mathrm{k=1}}^{\mathrm{N}} \varphi_{\mathrm{k}} \end{cases}$$

Cut-off frequency: f_{L} and f_{H}

$$\begin{cases} \frac{1}{f_{\mathrm{H}}} = 1.1\sqrt{\sum_{\mathrm{k=1}}^{\mathrm{N}} \frac{1}{f_{\mathrm{Hk}}^2}} \\ f_{\mathrm{L}} = 1.1\sqrt{\sum_{\mathrm{k=1}}^{\mathrm{N}} f_{\mathrm{Lk}}^2} \end{cases}$$

Bode plot of N-stage amplifier circuit

= Linear superposition of the Bode plot of each stage

多级放大电路波特图画法： 等于每一级电路波特图的线性叠加

Bode plot of 1st stage

+

Bode plot of 2nd stage

+

⋮

+

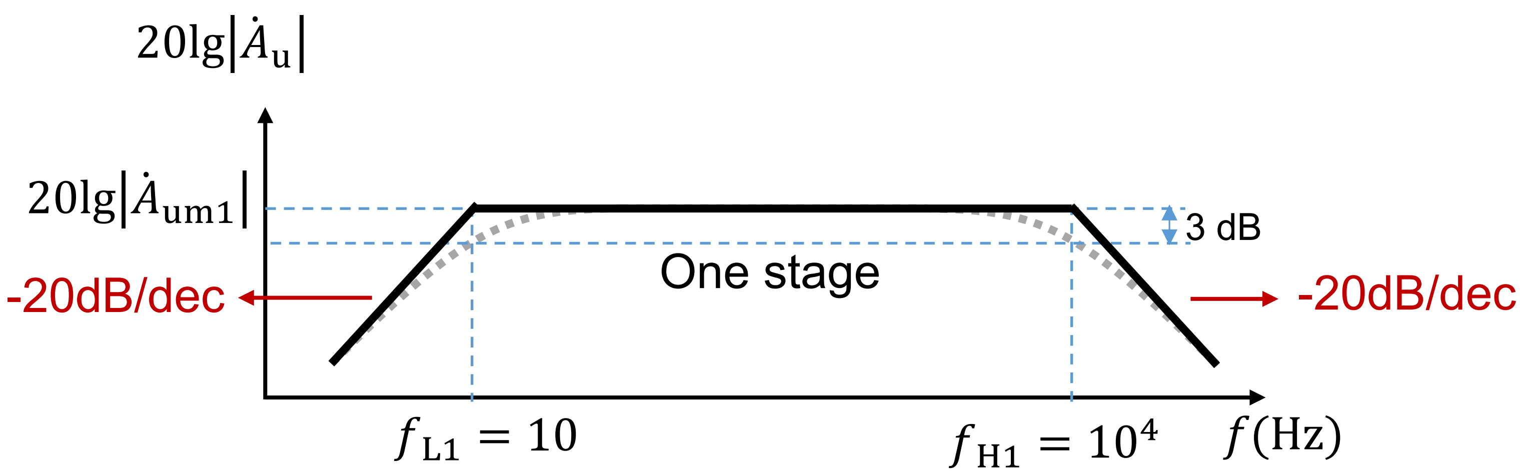
Bode plot of Nth stage

Q1: An amplifier circuit is made of two identical stages. The bode plot of each stage is shown in the figure below.

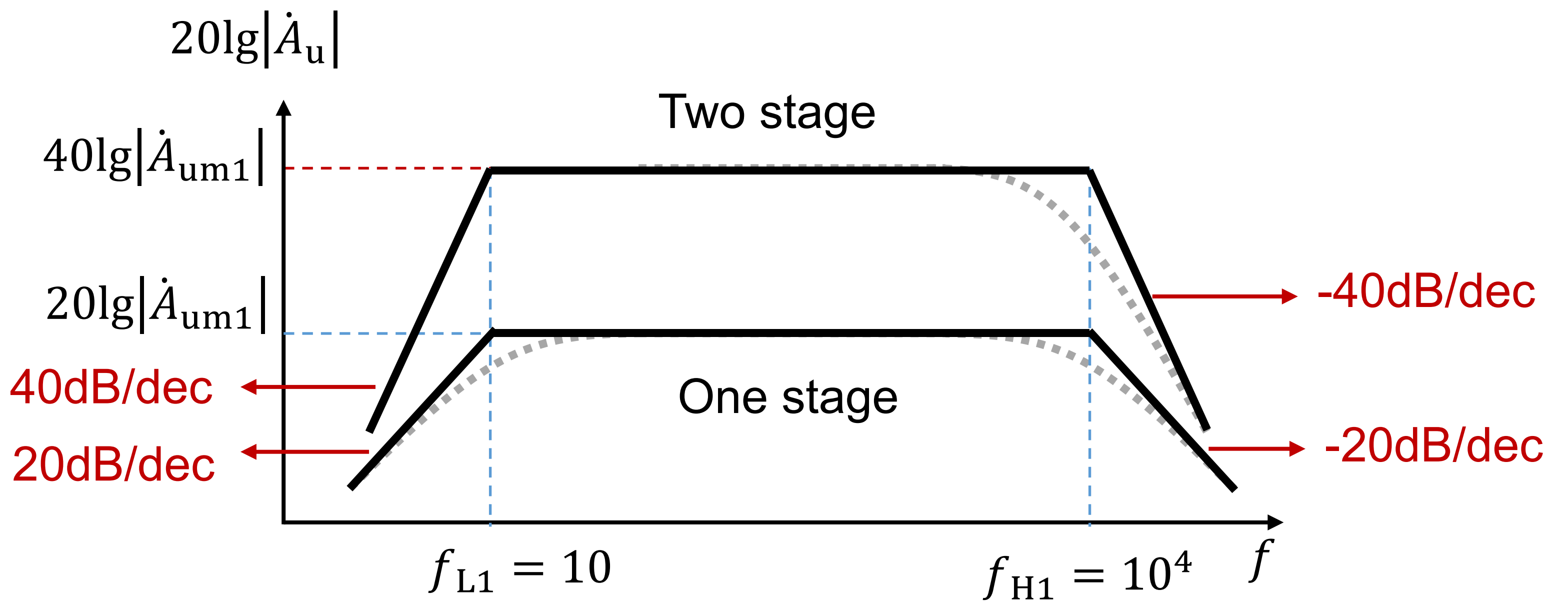
(1) Please draw the bode plot of the two-stage amplifier circuit;

(2) Lower and upper cut-off frequency of the two stage amplifier circuit;

(3) The voltage gain of the two-stage amplifier circuit for full wave band.



Answer (1):

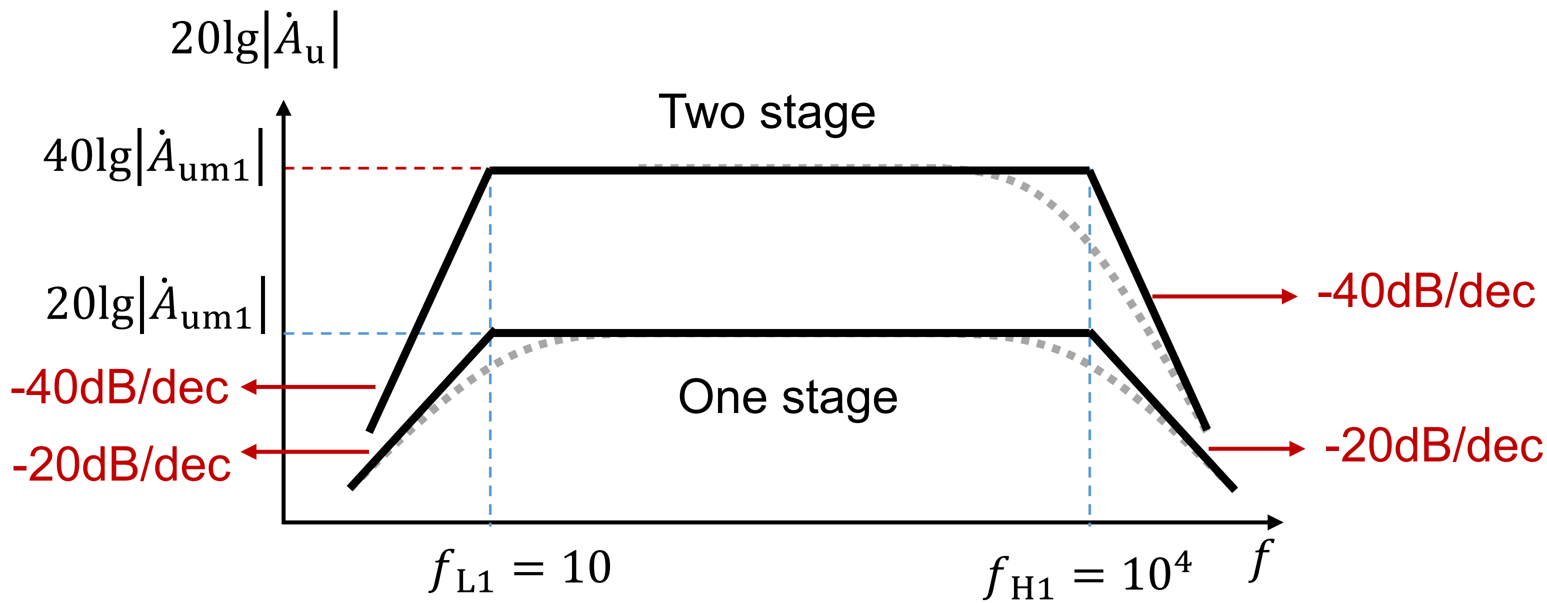


(2) Upper cut-off frequency for each stage: 10^4, 10^4 Hz

Upper cut-off frequency:

$$\frac{1}{f_H} \approx 1.1\sqrt{\sum_{k=1}^{N}\frac{1}{f_{Hk}^2}} = 1.1\sqrt{(\frac{1}{10^4})^2 + (\frac{1}{10^4})^2}; \quad f_H = 0.58 \times 10^4 \text{Hz}$$

Answer (2):

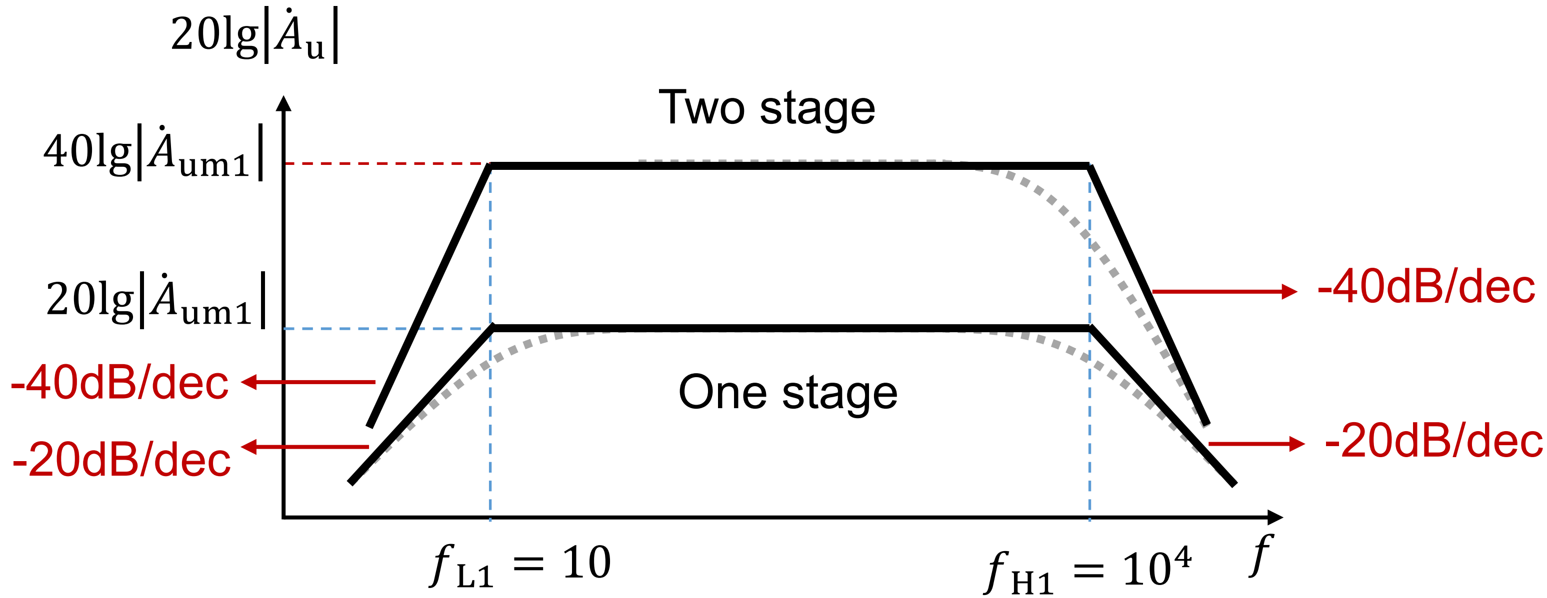


Lower cut-off frequency for each stage: 10 Hz, 10 Hz

Lower cut-off frequency:

$$f_L \approx 1.1\sqrt{\sum_{k=1}^{N} f_{Lk}^2} = 1.1\sqrt{10^2 + 10^2} = 15.6\text{ Hz}$$

Answer (3):



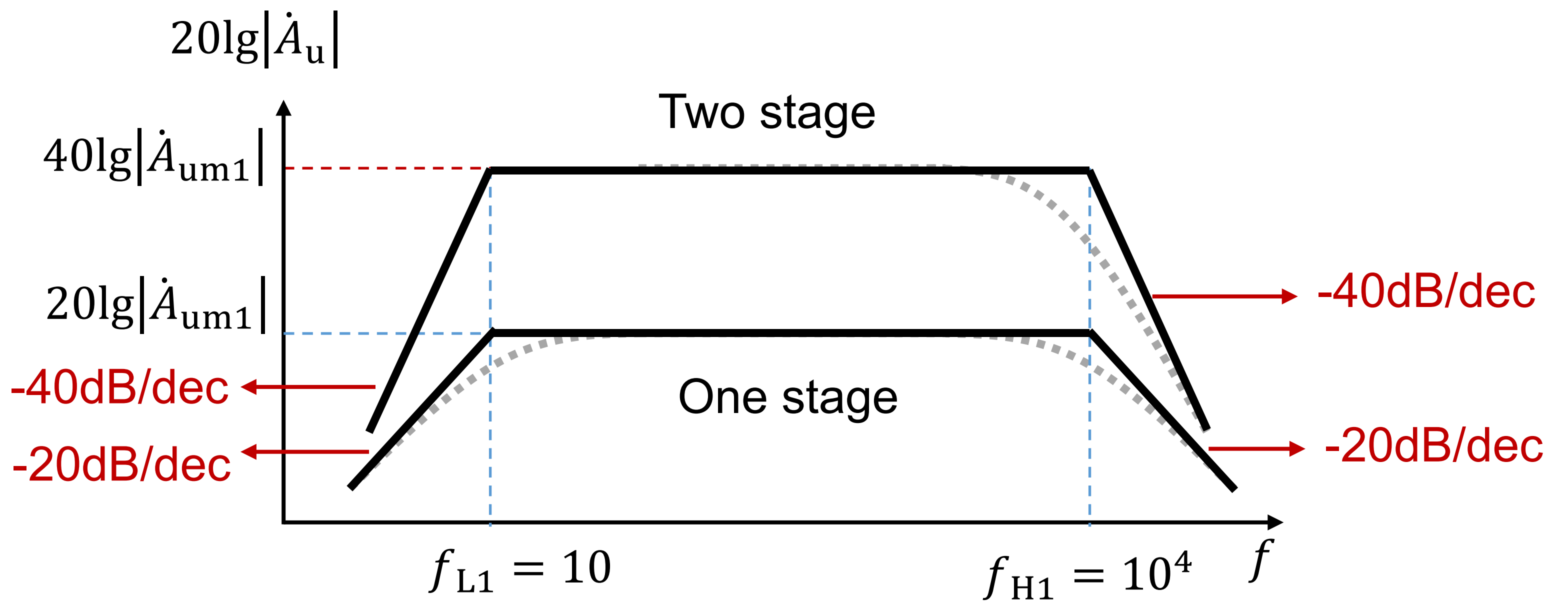
$$20\lg\dot{A}_{us}=$$

$$=40\lg\left|\dot{A}_{um1}\right|+20\lg\frac{1}{1+{f_{L1}}/{jf}}+20\lg\frac{1}{1+{f_{L2}}/{jf}}+20\lg\frac{1}{1+j{f}/{f_{H1}}}+20\lg\frac{1}{1+j{f}/{f_{H2}}}$$

$$=40\lg\left|\dot{A}_{um1}\right|+40\lg\frac{1}{1+{f_{L1}}/{jf}}+40\lg\frac{1}{1+j{f}/{f_{H1}}}$$

$$=40\lg\left|\dot{A}_{um1}\right|+40\lg\frac{1}{1+{10}/{jf}}+40\lg\frac{1}{1+j{f}/{10^4}}$$

Answer (3): another approach

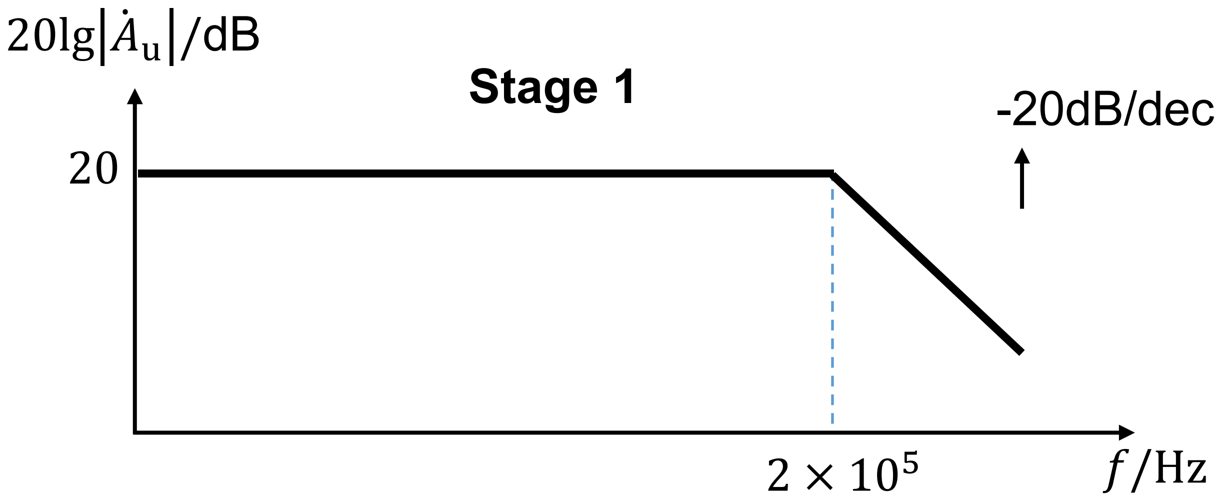


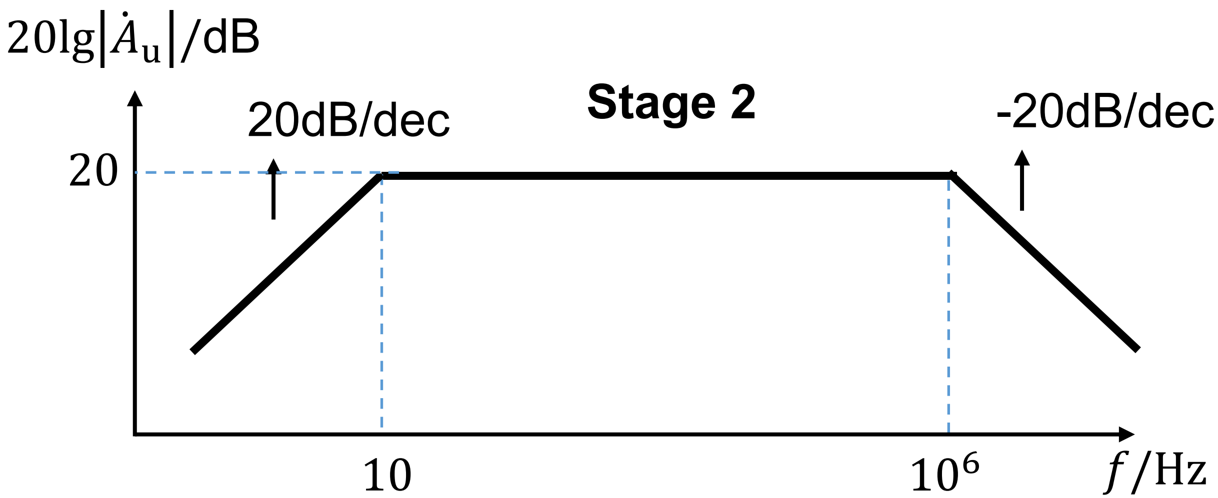
$$\dot{A}_{us} = \dot{A}_{um1}{}^2 \frac{1}{1 + {f_{L1}}/{jf}} \frac{1}{1 + {f_{L2}}/{jf}} \frac{1}{1 + j\,{f}/{f_{H1}}} \frac{1}{1 + j\,{f}/{f_{H2}}}$$

$$= \dot{A}_{um1}{}^2 \frac{1}{1 + {10}/{jf}} \frac{1}{1 + {10}/{jf}} \frac{1}{1 + j\,{f}/{10^4}} \frac{1}{1 + j\,{f}/{10^4}}$$

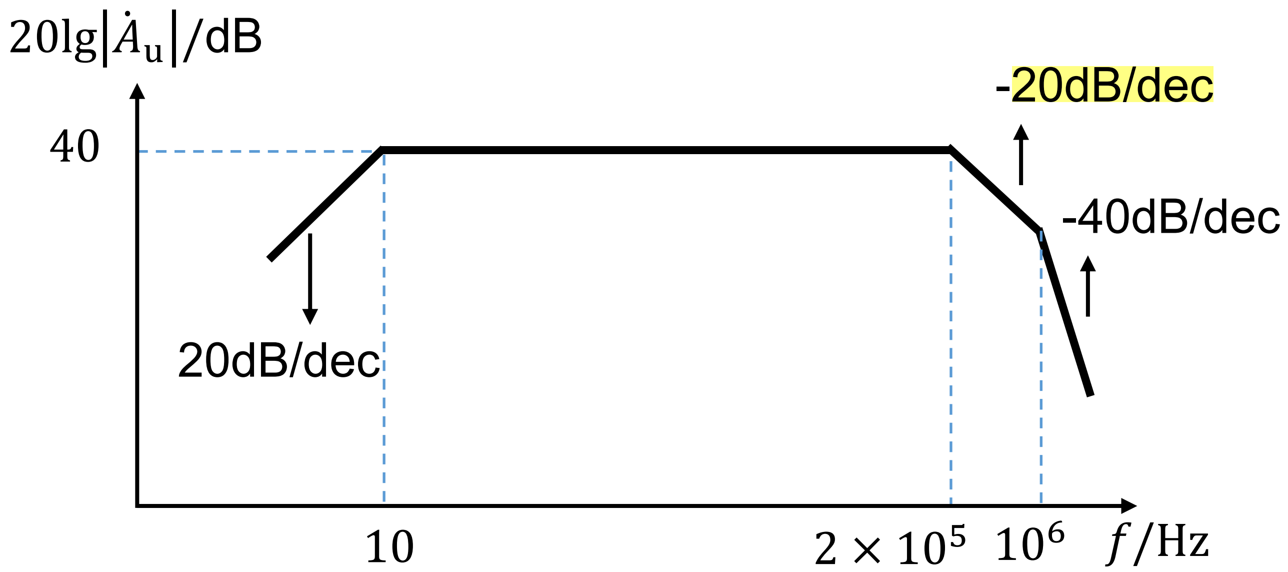
$$= \dot{A}_{um1}{}^2 \left(\frac{1}{1 + {10}/{jf}}\right)^2 \left(\frac{1}{1 + j\,{f}/{10^4}}\right)^2$$

Q2: An amplifier circuit is made of two stages. The bode plot of each stage is shown in the figure below. Please draw the bode plot of the two-stage amplifier circuit.

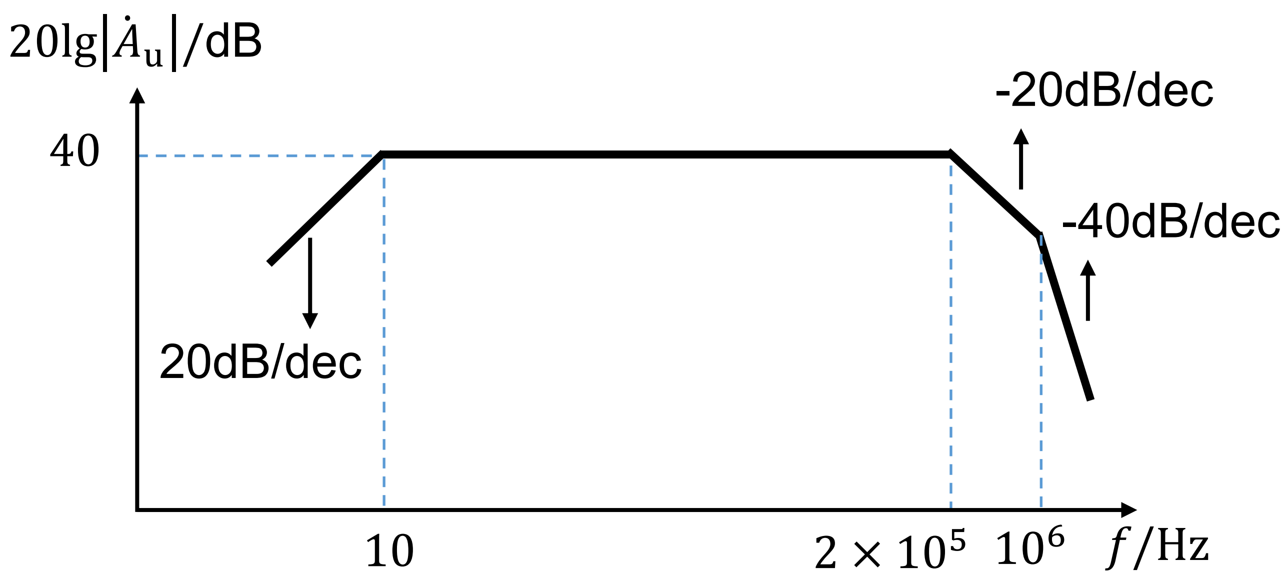


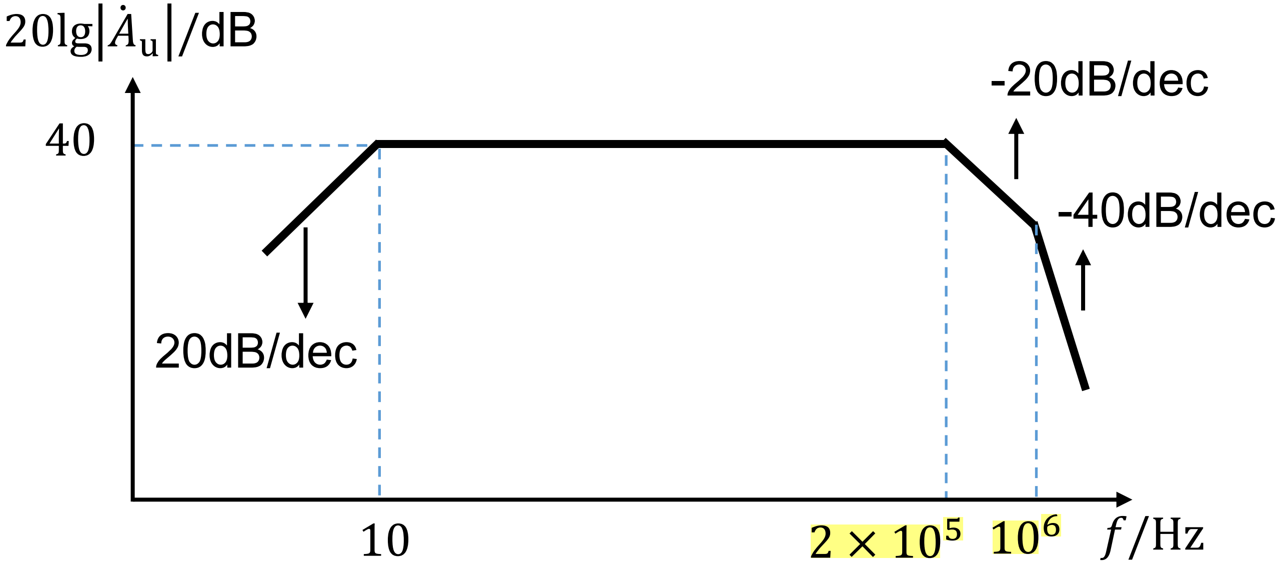


Answer:


$20\lg|\dot{A}_u|$/dB
-20dB/dec
40
-40dB/dec
20dB/dec
10
2×10^5
10^6
f/Hz

(2) lower and upper cut-off frequency?



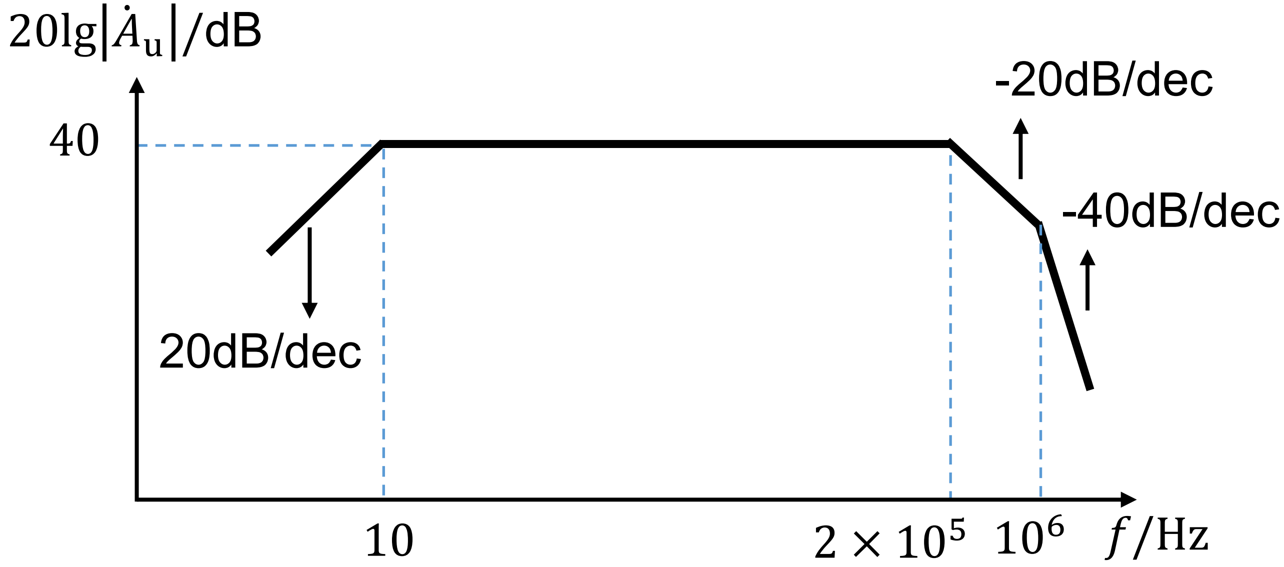


Lower cut-off frequency: 10 Hz

Upper cut-off frequency for each stage: 2×10^5, 10^6 Hz

Upper cut-off frequency:

$$\frac{1}{f_H} = 1.1\sqrt{\sum_{k=1}^{N}\frac{1}{f_{Hk}^2}} = 1.1\sqrt{\left(\frac{1}{2\times10^5}\right)^2 + \left(\frac{1}{10^6}\right)^2} \qquad f_H = 1.78\times10^5\ \mathrm{Hz}$$



(3) Expression of voltage gain for full wave band

$$\dot{A}_{us} = \dot{A}_{usm}\frac{1}{1+{f_L}/{jf}}\frac{1}{1+j{f}/{f_{H1}}}\frac{1}{1+j{f}/{f_{H2}}}$$

$$= 100\frac{1}{1+{10}/{jf}}\frac{1}{1+j{f}/{2\times10^5}}\frac{1}{1+j{f}/{10^6}}$$

Homework5: refer to black board

Deadline: 2025-05-07 11:59pm